Liver-mediated DDI study

Key results

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1.1. Rates and effect sizes

Biomarker	Control	Treatment	Effect size	Effect size	p-value	range
	mL/min/100mL	mL/min/100mL	mL/min/100mL	%		%
khe	13.1 +/- 4.3	0.8 +/- 0.3	-12.2 +/- 4.0	-93.2 +/- 1.7	0.0006	8.2
kbh	2.1 +/- 0.4	0.7 +/- 0.3	-1.4 +/- 0.4	-64.9 +/- 19.0	0.0003	82.7

Table 1.1: Primary outcomes for the drug rifampicin.

Biomarker	Control	Treatment	Effect size	Effect size	p-value	range
	mL/min/100mL	mL/min/100mL	mL/min/100mL	%		%
khe	15.8 +/- 5.8	3.6 +/- 1.5	-12.3 +/- 4.6	-77.3 +/- 5.9	0.0035	20.6
kbh	1.6 +/- 0.5	0.8 +/- 0.4	-0.8 +/- 0.3	-47.6 +/- 17.6	0.0062	51.7

Table 1.2: Primary outcomes for the drug ciclosporin.

Biomarker	Control	Treatment	Effect size	Effect size	p-value	range
	mL/min/100mL	mL/min/100mL	mL/min/100mL	%		%
khe	21.2 +/- 3.7	17.1 +/- 2.3	-4.2 +/- 2.6	-18.2 +/- 9.7	0.025	32.0
kbh	1.9 +/- 0.6	1.6 +/- 0.3	-0.3 +/- 0.5	-7.0 +/- 24.1	0.2859	79.5

Table 1.3: Primary outcomes for the drug metformin.

Biomarker	Control	Treatment	Effect size	Effect size	p-value	range
	mL/min/100mL	mL/min/100mL	mL/min/100mL	%		%
khe	3.0 +/- 2.2	1.2 +/- 0.9	-1.8 +/- 2.3	-39.1 +/- 60.2	0.2569	105.8
kbh	2.2 +/- 1.8	1.3 +/- 2.3	-0.9 +/- 2.6	-38.7 +/- 80.5	0.5734	138.5

Table 1.4: Primary outcomes for the drug rifampicin_clinical.

1.2. Changes during the visit

Biomarker	Visit	Start	End	Change	p-value	range
		mL/min/100mL	mL/min/100mL	%		%
khe	Control	11.5 +/- 2.5	14.7 +/- 6.7	25.4 +/- 38.2	0.2337	140.8
khe	Treatment	0.8 +/- 0.3	1.0 +/- 0.2	33.1 +/- 26.3	0.0427	117.9

Table 1.5: Secondary outcomes for the drug rifampicin.

Biomarker	Visit	Start	End	Change	p-value	range
		mL/min/100mL	mL/min/100mL	%		%
khe	Control	15.2 +/- 5.5	16.4 +/- 7.9	12.8 +/- 35.5	0.5126	124.3
khe	Treatment	3.6 +/- 1.5	7.9 +/- 2.2	148.6 +/- 58.6	0.0042	212.9

Table 1.6: Secondary outcomes for the drug ciclosporin.

Biomarker	Visit	Start	End	Change	p-value	range
		mL/min/100mL	mL/min/100mL	%		%
khe	Control	16.6 +/- 5.1	25.9 +/- 7.1	115.3 +/- 172.1	0.2462	585.2
khe	Treatment	17.1 +/- 2.3	27.4 +/- 6.4	57.7 +/- 21.7	0.0034	73.8

Table 1.7: Secondary outcomes for the drug metformin.

Biomarker	Visit	Start	End	Change	p-value	range
		mL/min/100mL	mL/min/100mL	%		%
khe	Control	2.7 +/- 2.1	3.3 +/- 2.6	32.1 +/- 55.6	0.3756	96.7
khe	Treatment	1.2 +/- 0.9	2.5 +/- 2.8	81.9 +/- 80.4	0.184	141.3

Table 1.8: Secondary outcomes for the drug rifampicin_clinical.

1.3. Constants

biomarker	units	value (95%CI)
Cardiac output	L/min	5.9 +/- 1.0
Heart-lung mean transit time	sec	17.0 +/- 2.3
Heart-lung dispersion	%	46.0 +/- 9.0
Organs blood mean transit time	sec	32.0 +/- 8.0
Organs extraction fraction	%	18.0 +/- 3.0
Organs extravascular mean transit time	min	6.7 +/- 1.6
Gut mean transit time	sec	44.0 +/- 7.9
Gut dispersion	%	74.0 +/- 5.0
Liver extracellular volume fraction	%	20.0 +/- 3.0

Table 1.9: Constants for the drug rifampicin.

biomarker	units	value (95%CI)
Cardiac output	L/min	8.0 +/- 2.0
Heart-lung mean transit time	sec	16.0 +/- 5.1
Heart-lung dispersion	%	53.0 +/- 11.0
Organs blood mean transit time	sec	29.0 +/- 7.0
Organs extraction fraction	%	16.0 +/- 3.0
Organs extravascular mean transit time	min	6.3 +/- 1.5
Gut mean transit time	sec	27.0 +/- 6.0
Gut dispersion	%	73.0 +/- 9.0
Liver extracellular volume fraction	%	26.0 +/- 3.0

Table 1.10: Constants for the drug ciclosporin.

biomarker	units	value (95%CI)
Cardiac output	L/min	8.2 +/- 3.0
Heart-lung mean transit time	sec	18.0 +/- 4.5
Heart-lung dispersion	%	53.0 +/- 16.0
Organs blood mean transit time	sec	29.0 +/- 7.0
Organs extraction fraction	%	19.0 +/- 2.0
Organs extravascular mean transit time	min	8.6 +/- 1.6
Gut mean transit time	sec	31.0 +/- 8.1
Gut dispersion	%	77.0 +/- 9.0
Liver extracellular volume fraction	%	28.0 +/- 3.0

Table 1.11: Constants for the drug metformin.

biomarker	units	value (95%CI)
Cardiac output	L/min	7.3 +/- 0.8
Heart-lung mean transit time	sec	17.0 +/- 2.7
Heart-lung dispersion	%	44.0 +/- 17.0
Organs blood mean transit time	sec	30.0 +/- 8.0
Organs extraction fraction	%	17.0 +/- 2.0
Organs extravascular mean transit time	min	5.4 +/- 2.5
Gut mean transit time	sec	54.0 +/- 5.9
Gut dispersion	%	81.0 +/- 7.0
Liver extracellular volume fraction	%	26.0 +/- 8.0

Table 1.12: Constants for the drug rifampicin_clinical.

1.4. Effect plots

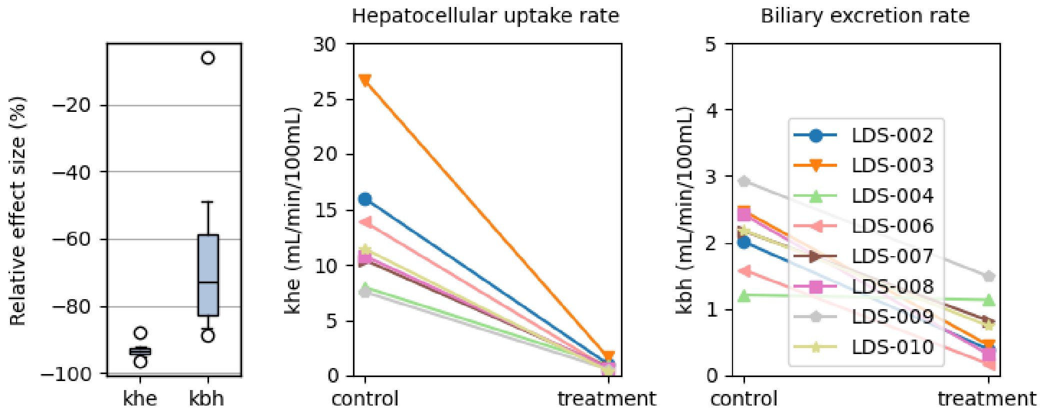

Figure 1.1: Effect of the drug rifampicin

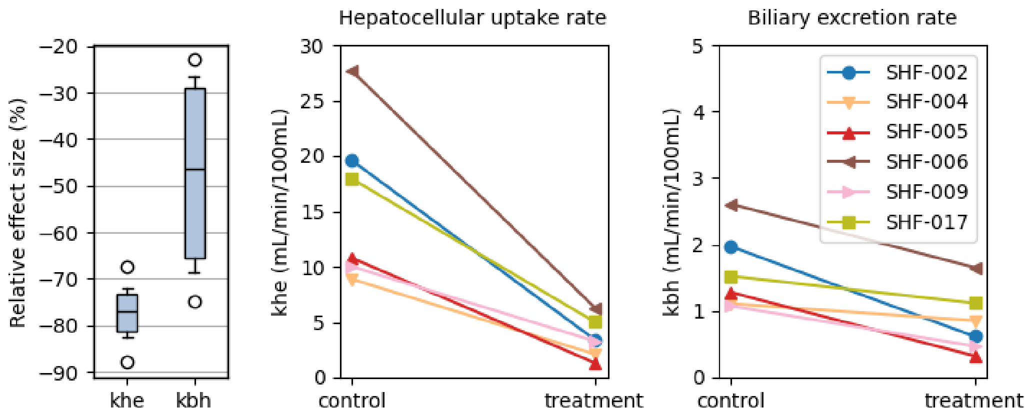

Figure 1.2: Effect of the drug ciclosporin

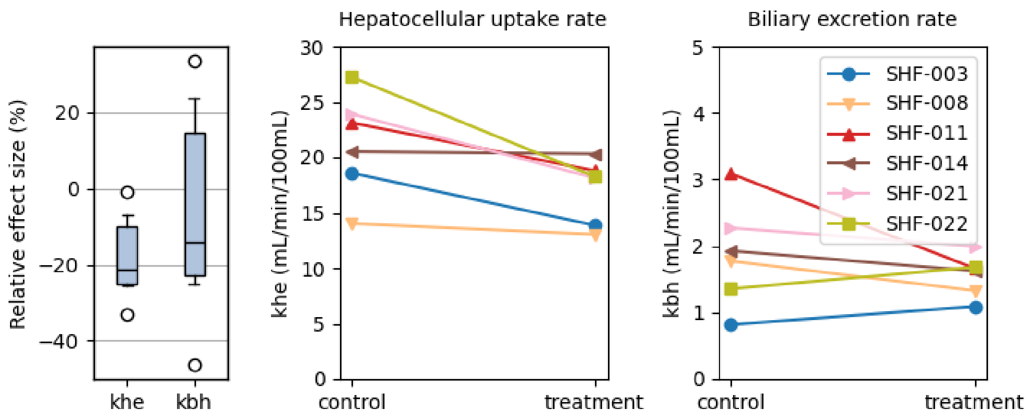

Figure 1.3: Effect of the drug metformin

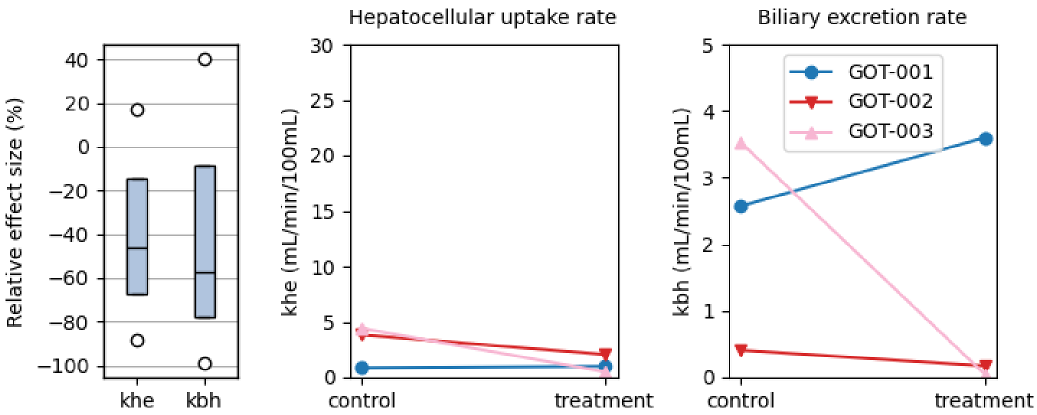

Figure 1.4: Effect of the drug rifampicin_clinical

1.5. Diurnal variations

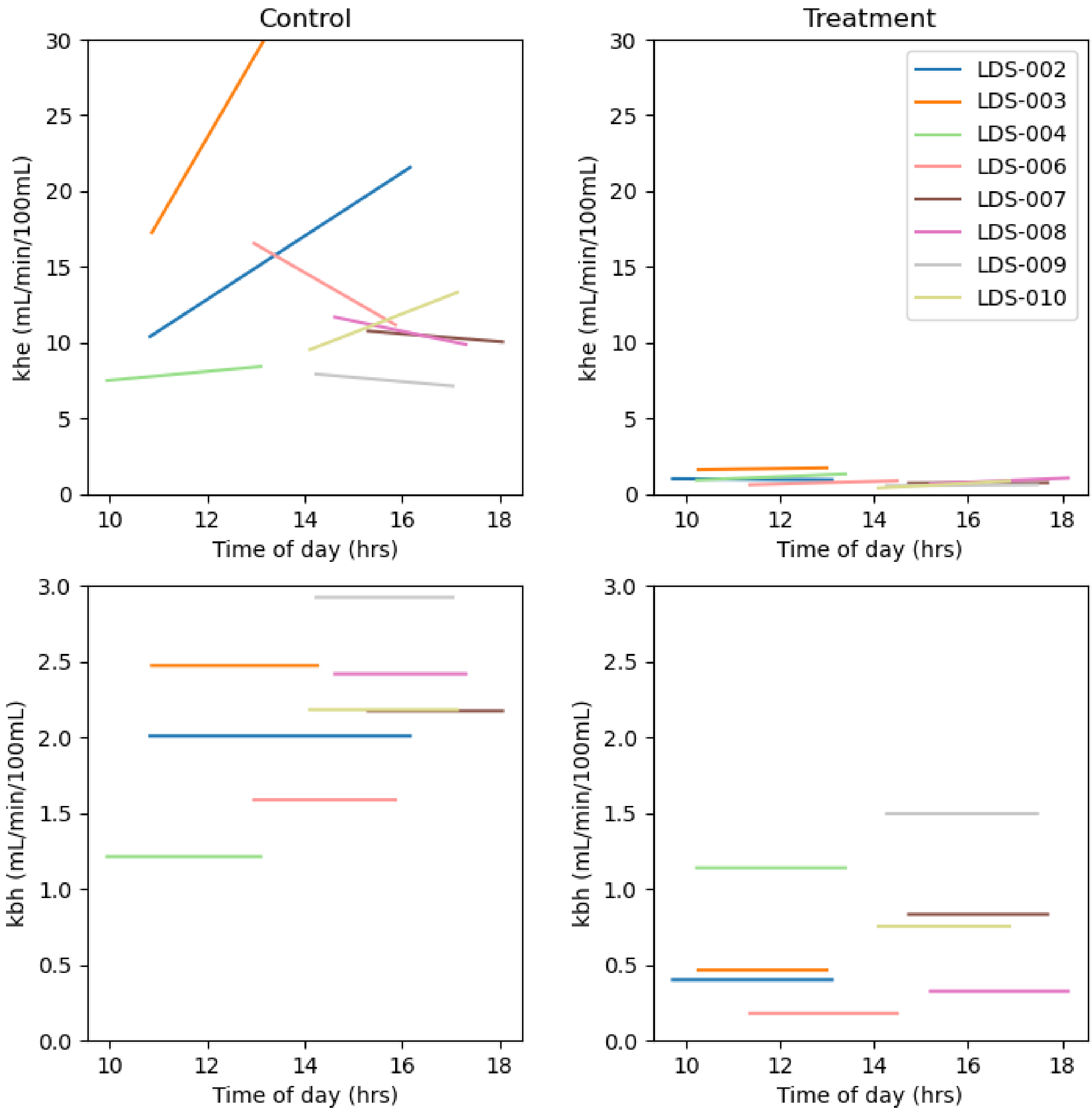

Figure 1.5: Effect of the drug rifampicin

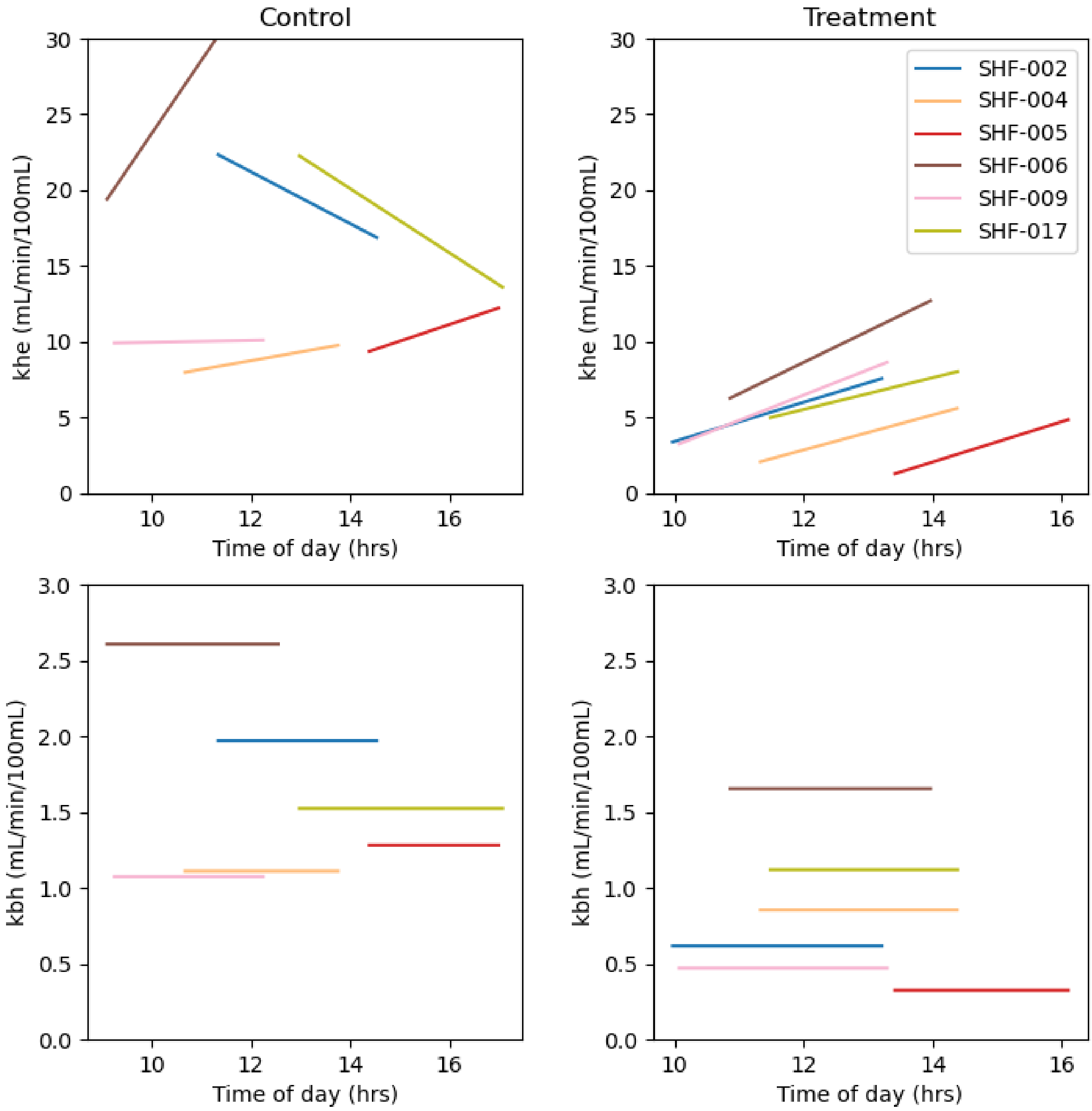

Figure 1.6: Effect of the drug ciclosporin

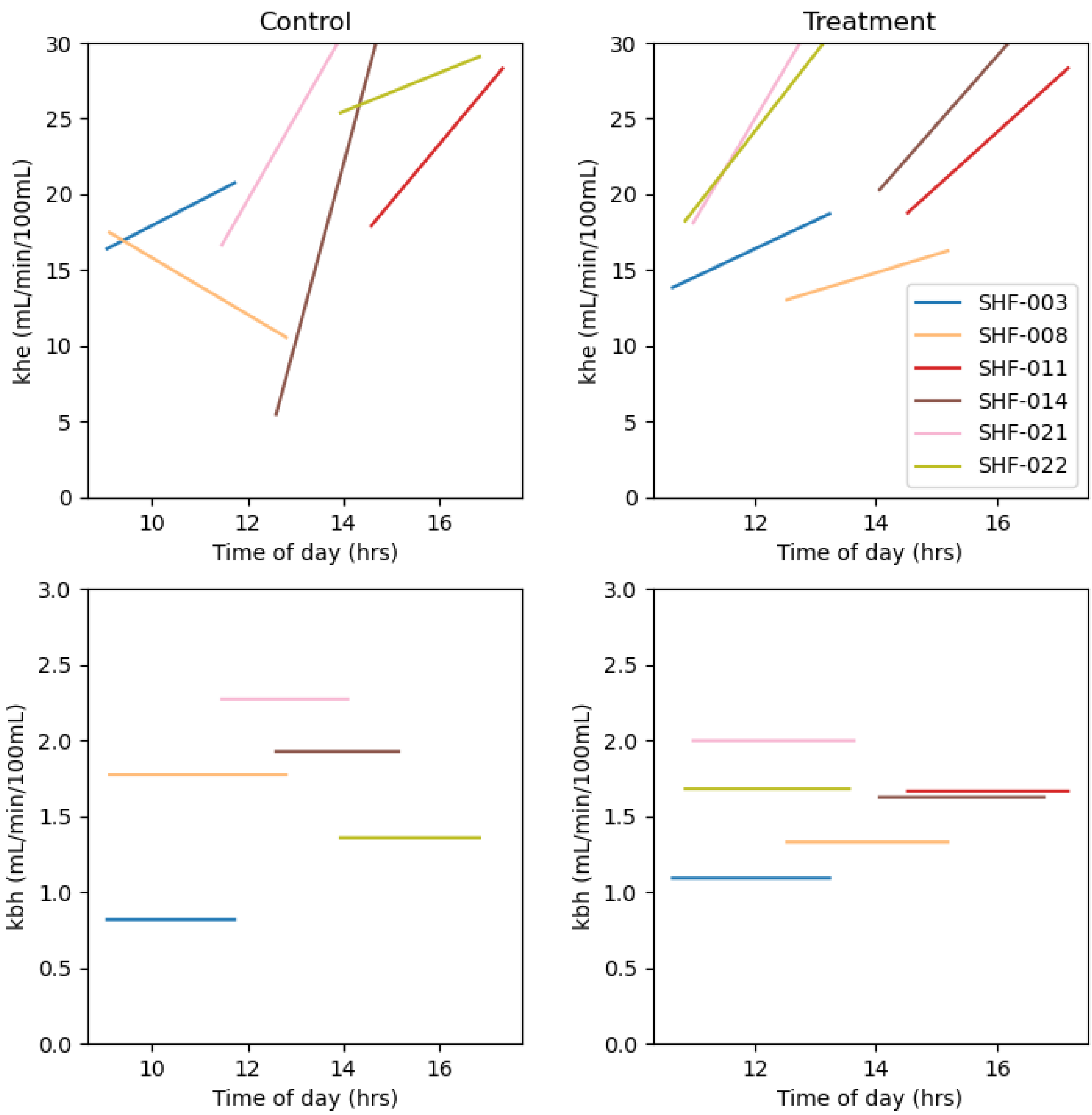

Figure 1.7: Effect of the drug metformin

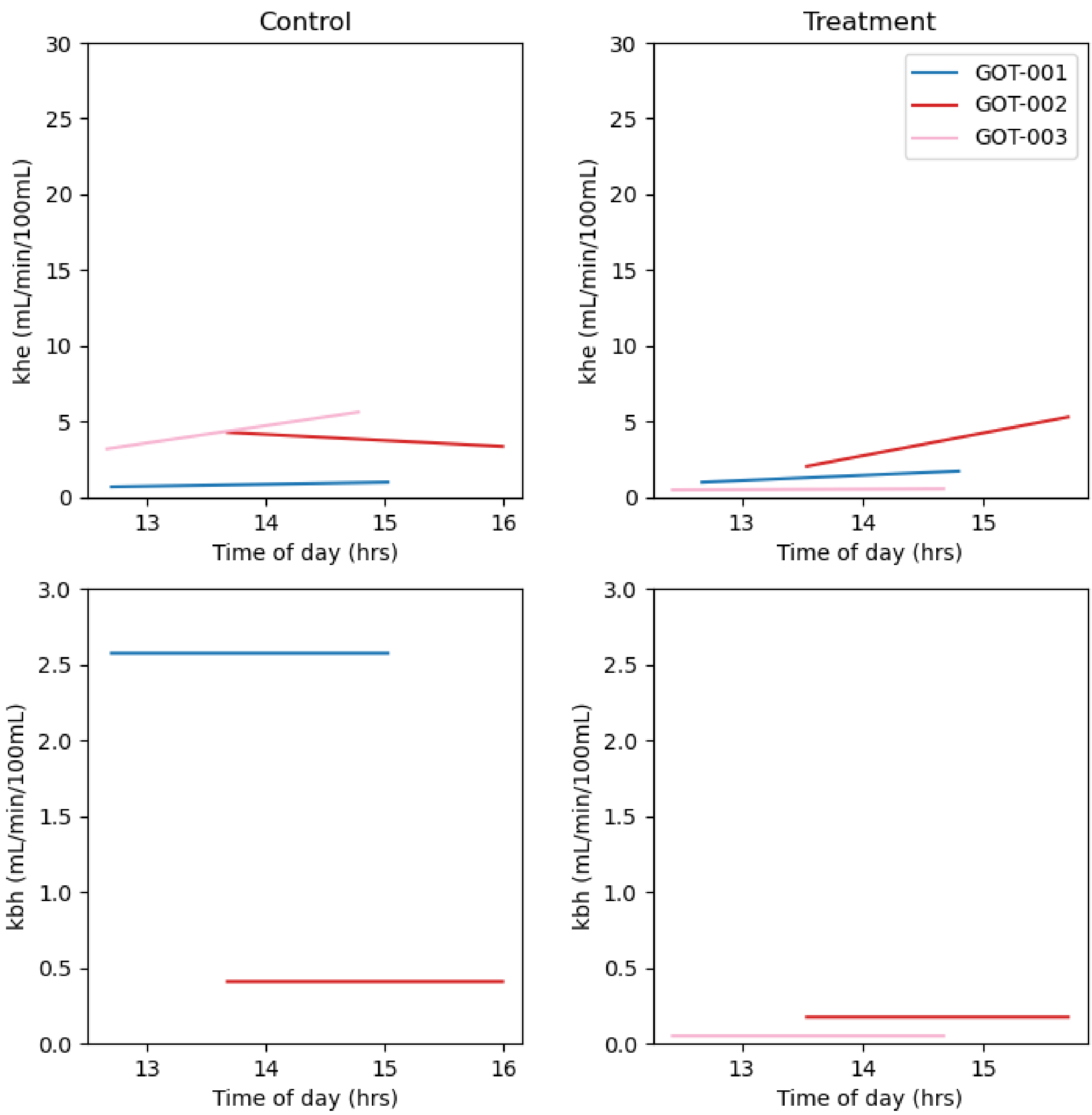

Figure 1.8: Effect of the drug rifampicin_clinical

1.6. Model fits

1.6.1. rifampicin

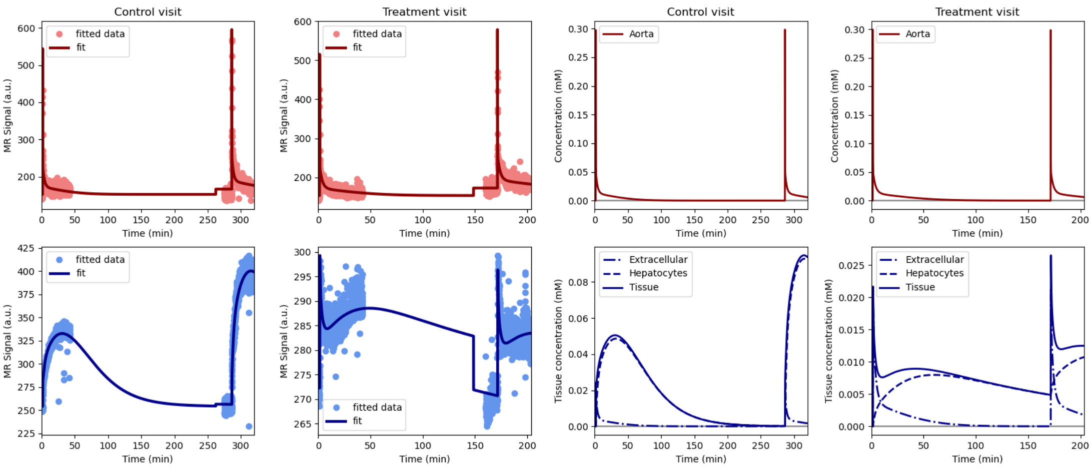

Figure 1.9: Volunteer LDS-002.png, drug rifampicin

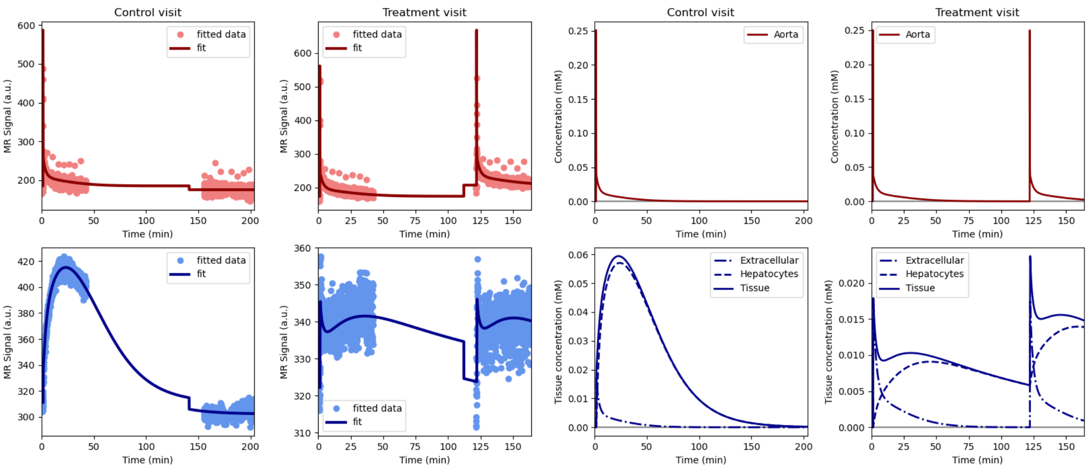

Figure 1.10: Volunteer LDS-003.png, drug rifampicin

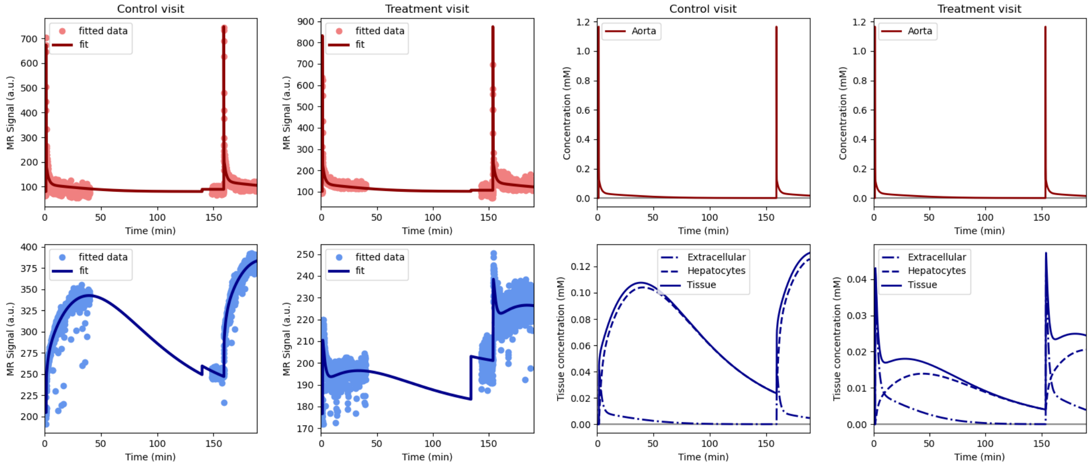

Figure 1.11: Volunteer LDS-004.png, drug rifampicin

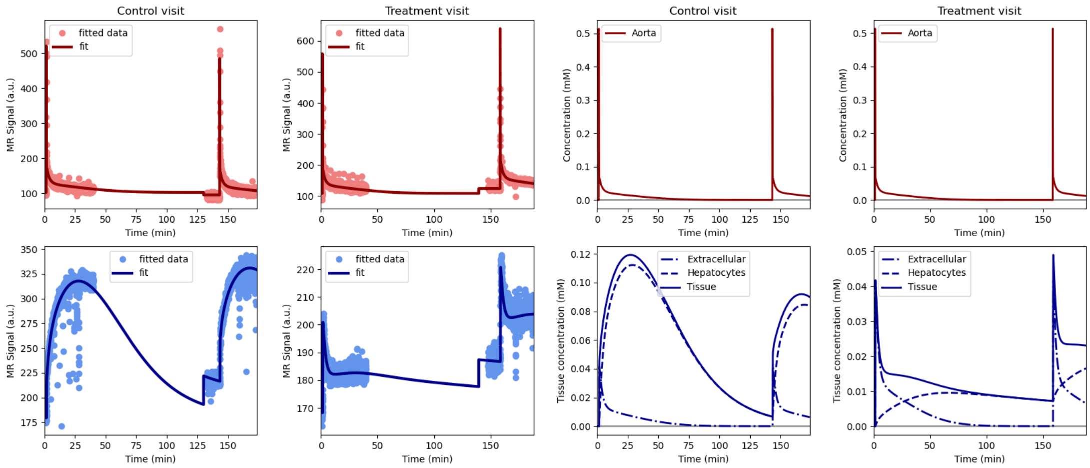

Figure 1.12: Volunteer LDS-006.png, drug rifampicin

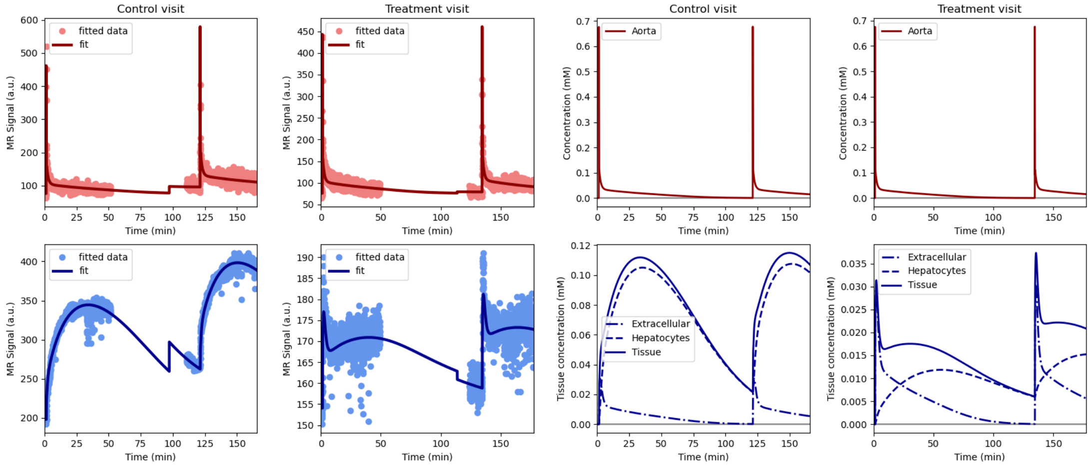

Figure 1.13: Volunteer LDS-007.png, drug rifampicin

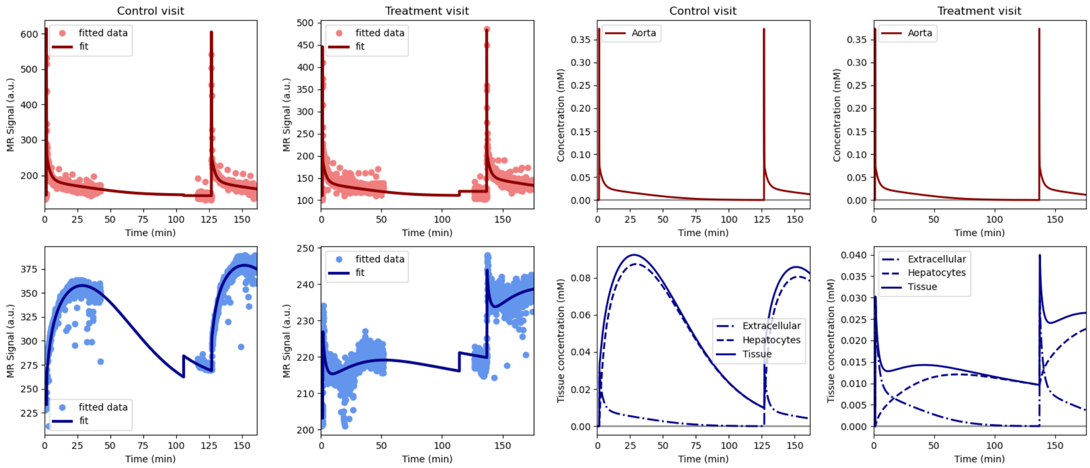

Figure 1.14: Volunteer LDS-008.png, drug rifampicin

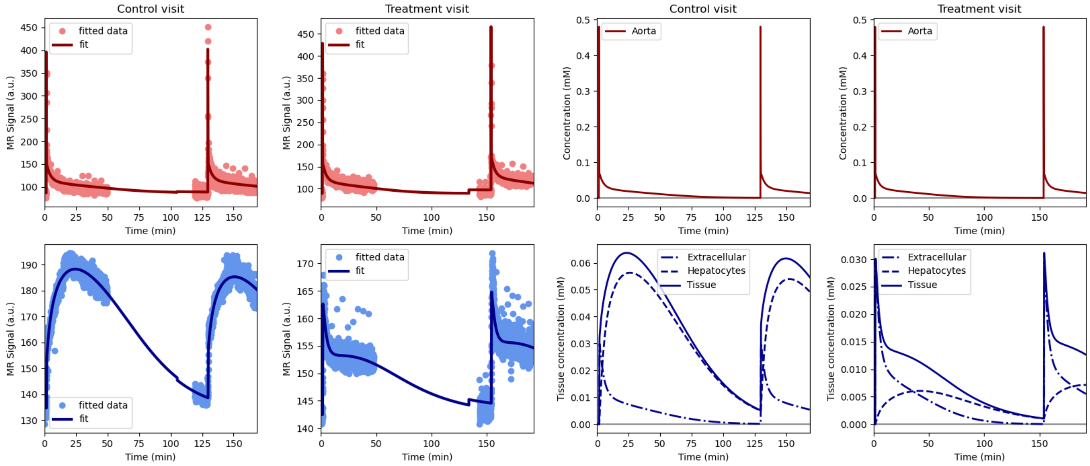

Figure 1.15: Volunteer LDS-009.png, drug rifampicin

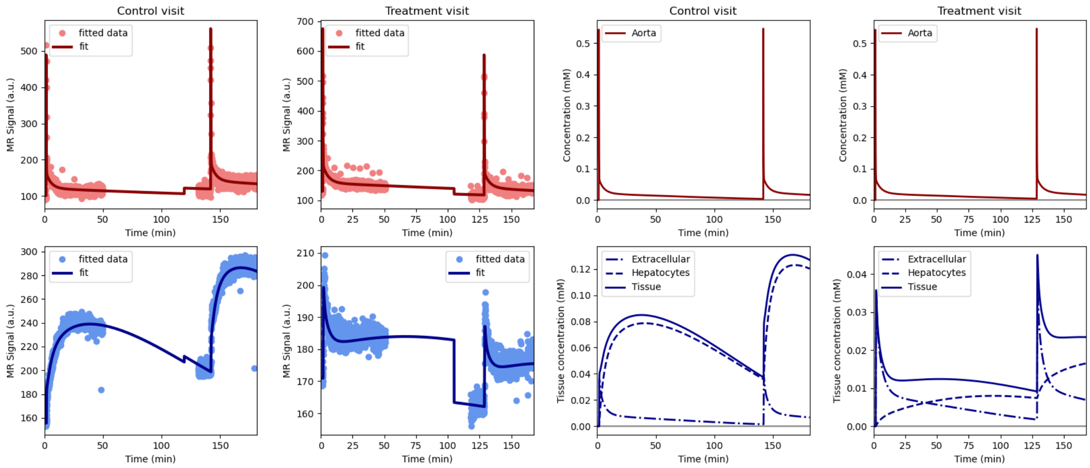

Figure 1.16: Volunteer LDS-010.png, drug rifampicin

1.6.2. ciclosporin

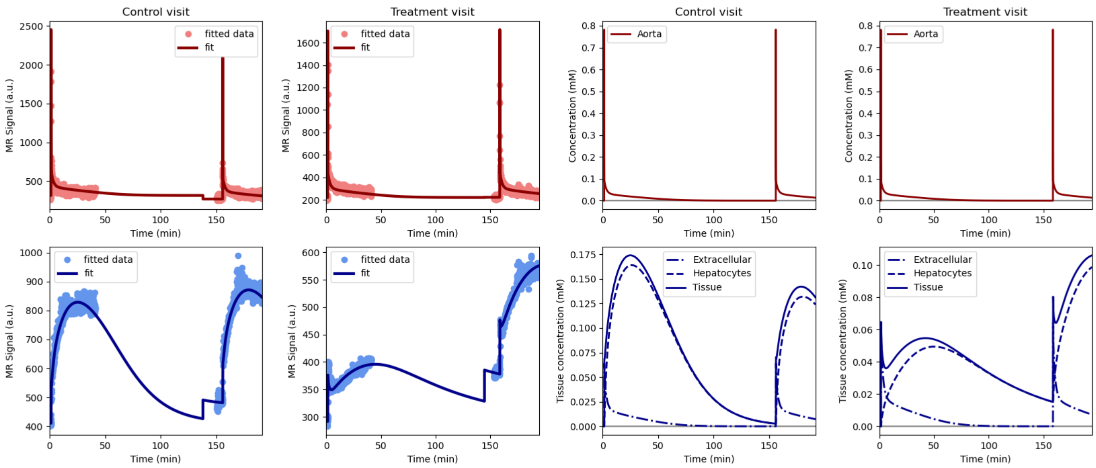

Figure 1.17: Volunteer SHF-002.png, drug ciclosporin

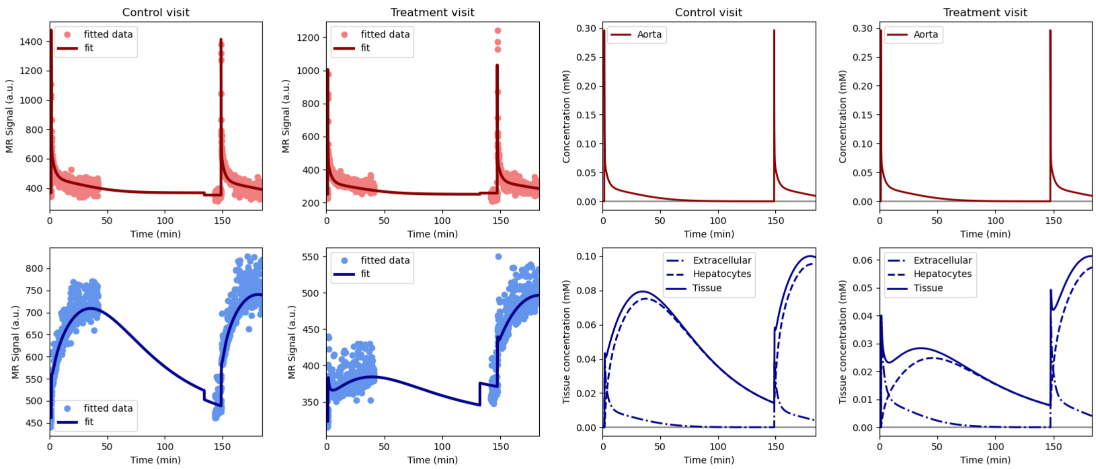

Figure 1.18: Volunteer SHF-004.png, drug ciclosporin

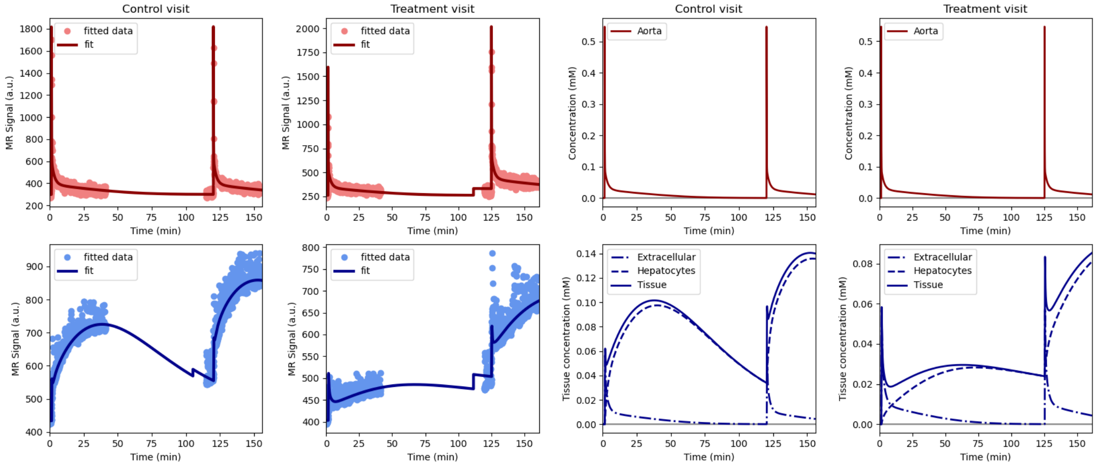

Figure 1.19: Volunteer SHF-005.png, drug ciclosporin

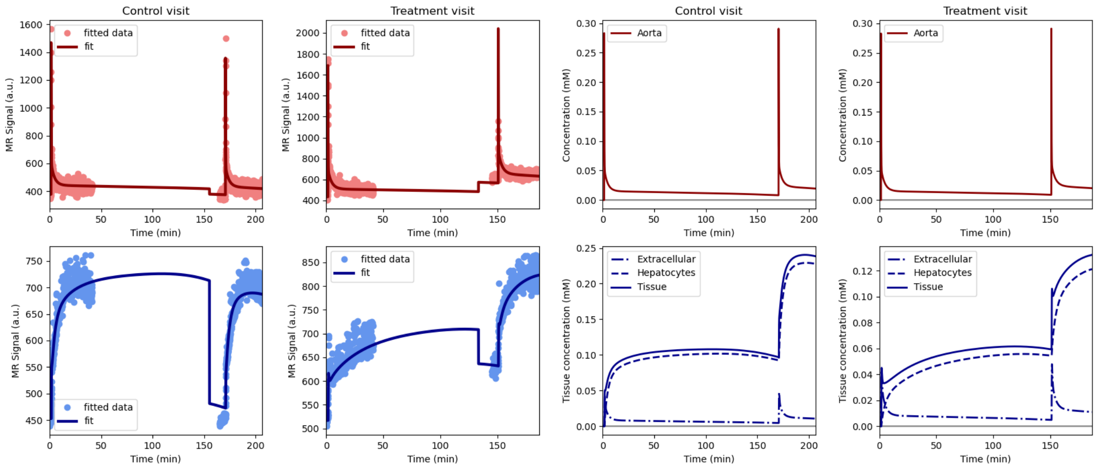

Figure 1.20: Volunteer SHF-006.png, drug ciclosporin

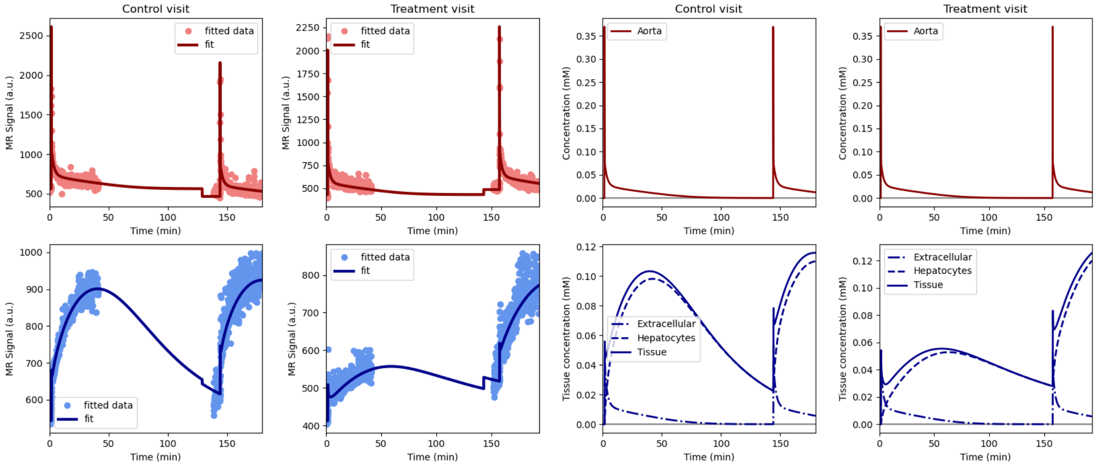

Figure 1.21: Volunteer SHF-009.png, drug ciclosporin

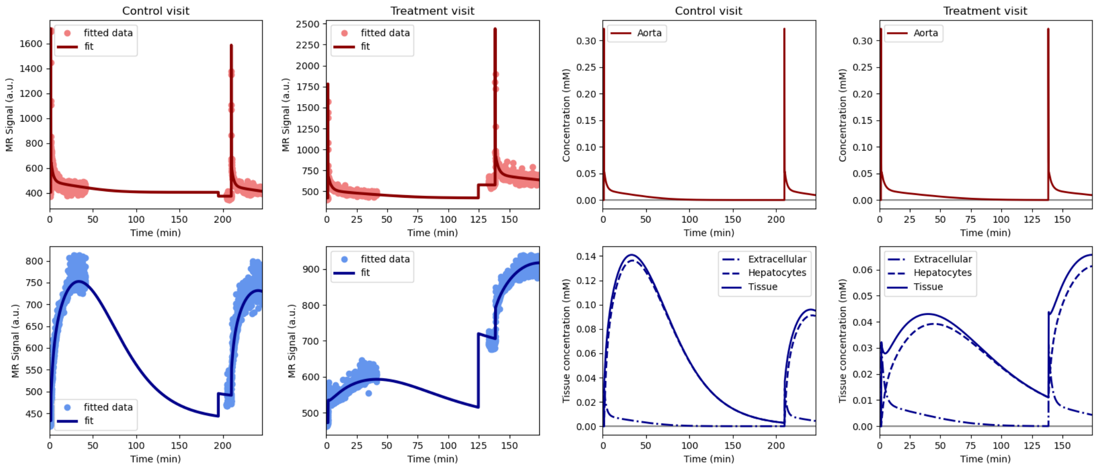

Figure 1.22: Volunteer SHF-017.png, drug ciclosporin

1.6.3. metformin

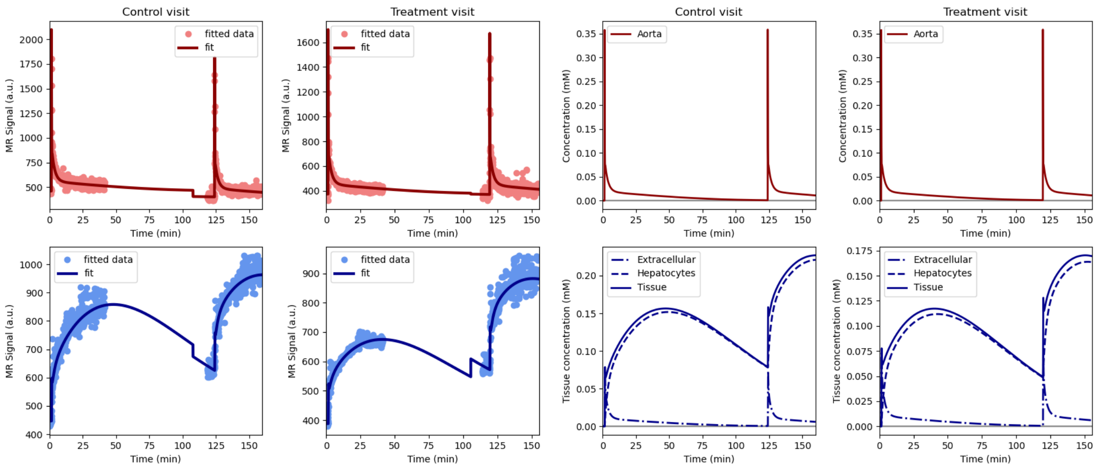

Figure 1.23: Volunteer SHF-003.png, drug metformin

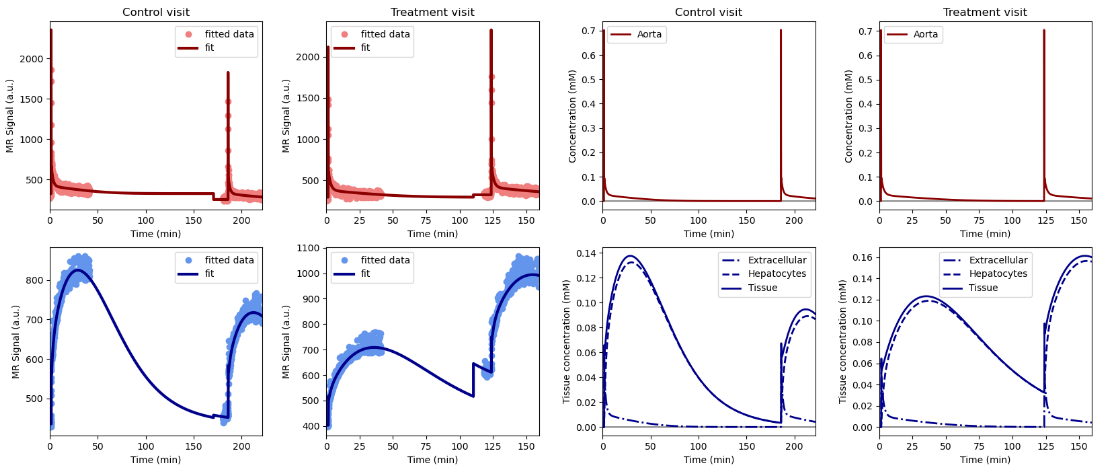

Figure 1.24: Volunteer SHF-008.png, drug metformin

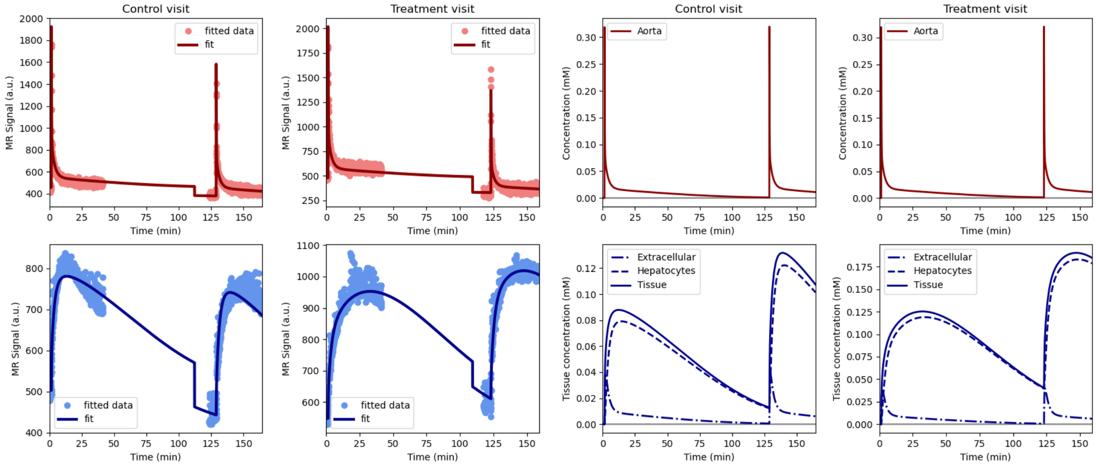

Figure 1.25: Volunteer SHF-011.png, drug metformin

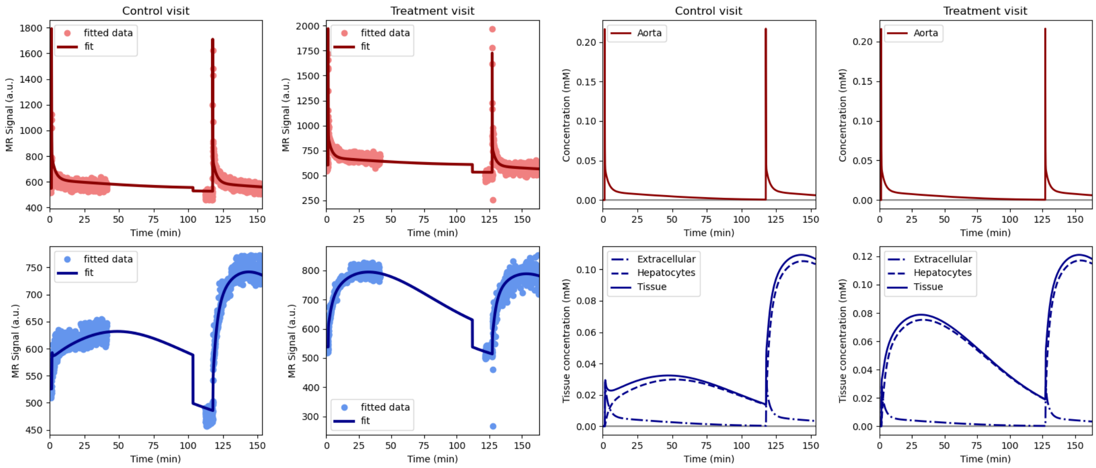

Figure 1.26: Volunteer SHF-014.png, drug metformin

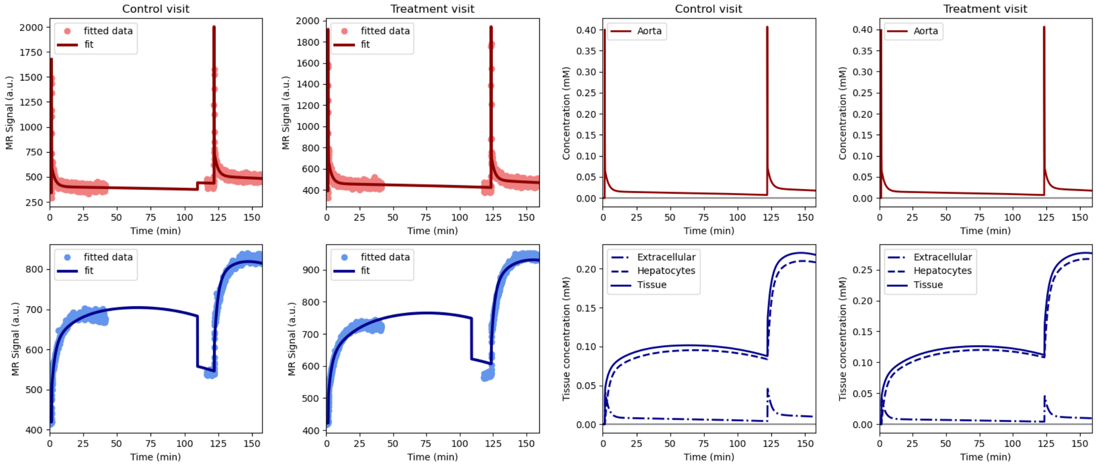

Figure 1.27: Volunteer SHF-021.png, drug metformin

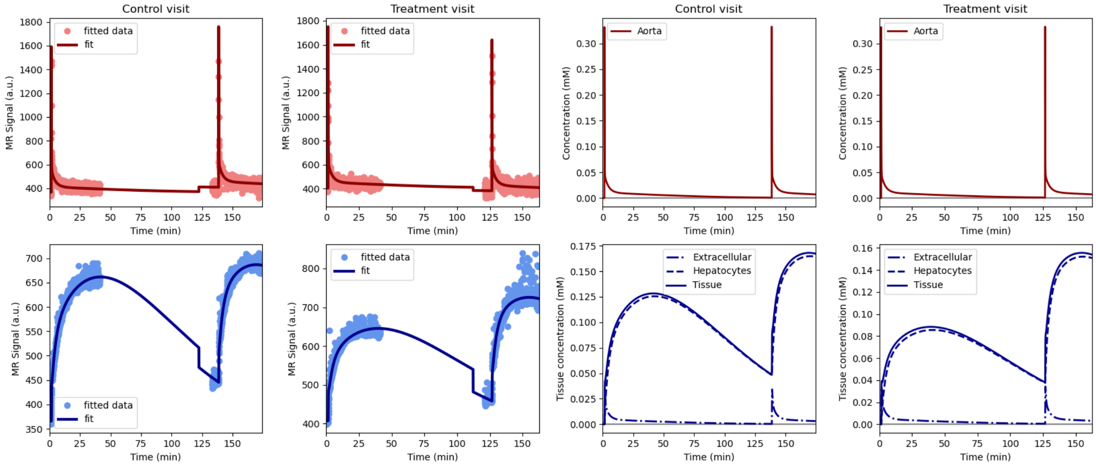

Figure 1.28: Volunteer SHF-022.png, drug metformin

1.6.4. rifampicin_clinical

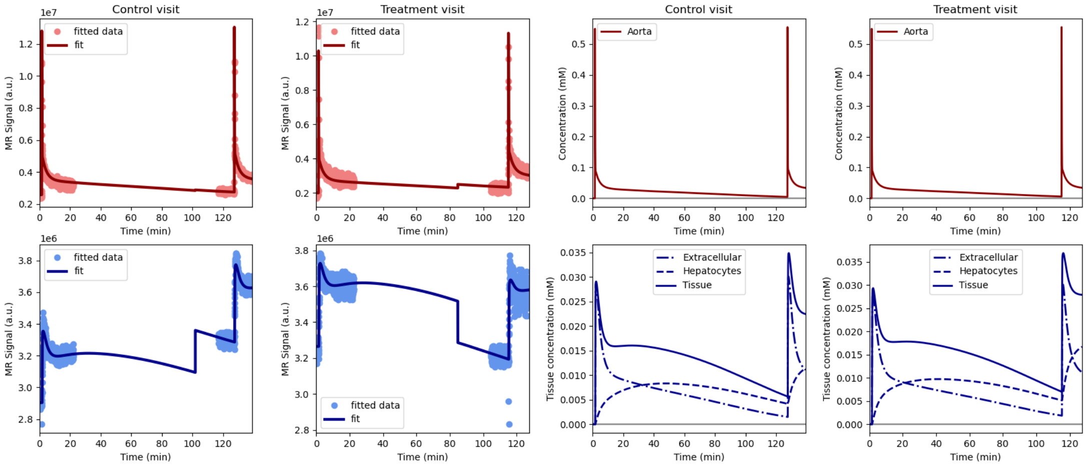

Figure 1.29: Volunteer GOT-001.png, drug rifampicin_clinical

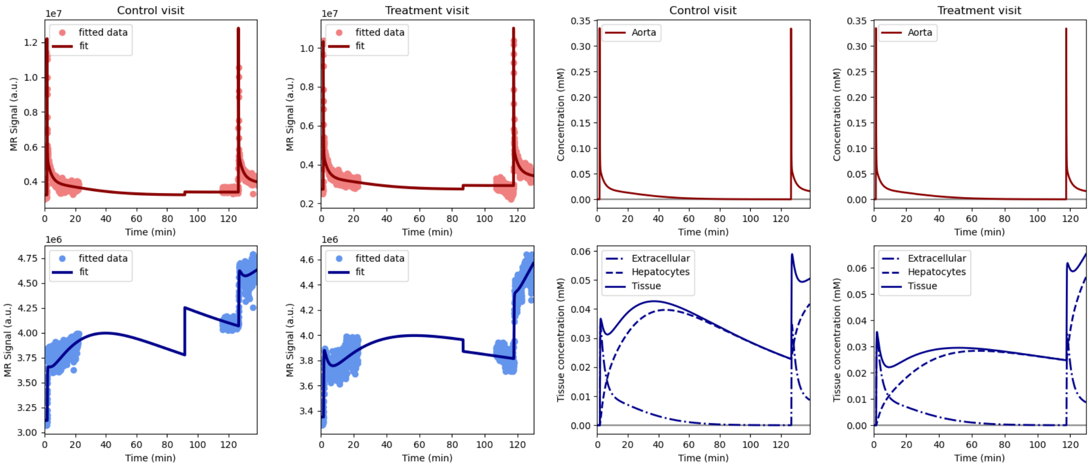

Figure 1.30: Volunteer GOT-002.png, drug rifampicin_clinical

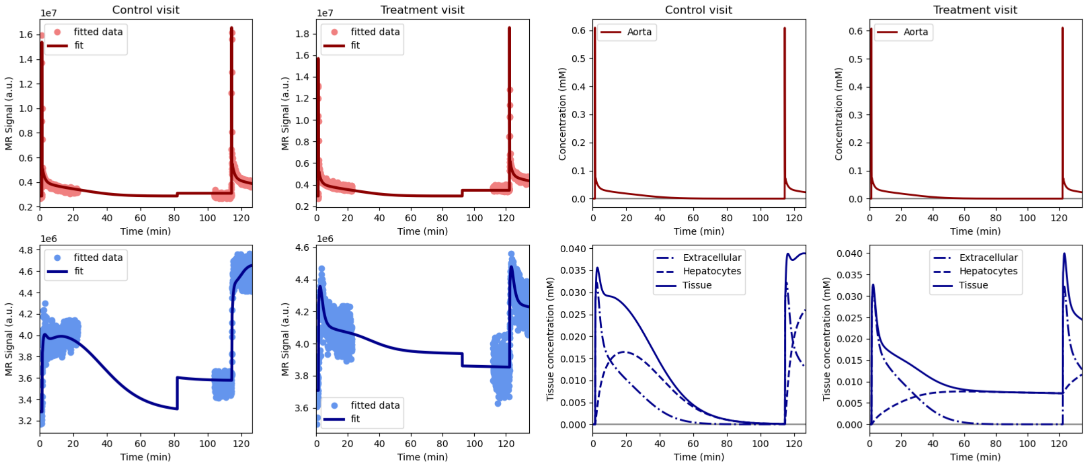

Figure 1.31: Volunteer GOT-003.png, drug rifampicin_clinical

2

One scan

2.1. Rates and effect sizes

Biomarker	Control	Treatment	Effect size	Effect size	p-value	range
	mL/min/100mL	mL/min/100mL	mL/min/100mL	%		%
khe	11.9 +/- 2.7	0.8 +/- 0.2	-11.1 +/- 2.6	-92.8 +/- 1.7	0.0001	7.7
kbh	2.0 +/- 0.4	0.4 +/- 0.3	-1.6 +/- 0.4	-78.7 +/- 16.6	0.0002	63.5

Table 2.1: Primary outcomes for the drug rifampicin.

Biomarker	Control	Treatment	Effect size	Effect size	p-value	range
	mL/min/100mL	mL/min/100mL	mL/min/100mL	%		%
khe	15.6 +/- 5.0	3.3 +/- 1.0	-12.4 +/- 4.5	-78.0 +/- 6.0	0.003	19.0
kbh	1.4 +/- 0.3	0.3 +/- 0.2	-1.1 +/- 0.3	-80.4 +/- 15.3	0.0008	45.8

Table 2.2: Primary outcomes for the drug ciclosporin.

Biomarker	Control	Treatment	Effect size	Effect size	p-value	range
	mL/min/100mL	mL/min/100mL	mL/min/100mL	%		%
khe	18.9 +/- 5.2	19.8 +/- 4.2	0.9 +/- 7.7	29.6 +/- 84.1	0.822	268.9
kbh	2.1 +/- 1.0	1.7 +/- 0.6	-0.3 +/- 0.8	-0.8 +/- 34.0	0.4407	103.2

Table 2.3: Primary outcomes for the drug metformin.

Biomarker	Control	Treatment	Effect size	Effect size	p-value	range
	mL/min/100mL	mL/min/100mL	mL/min/100mL	%		%
khe	2.7 +/- 2.4	1.3 +/- 1.0	-1.4 +/- 1.8	-28.8 +/- 73.2	0.2553	121.4
kbh	1.8 +/- 0.6	1.6 +/- 2.1	-0.2 +/- 1.5	-23.7 +/- 85.8	0.8211	151.6

Table 2.4: Primary outcomes for the drug rifampicin_clinical.

2.2. Constants

biomarker	units	value (95%CI)
Cardiac output	L/min	5.5 +/- 1.3
Heart-lung mean transit time	sec	16.0 +/- 1.8
Heart-lung dispersion	%	42.0 +/- 6.0
Organs blood mean transit time	sec	34.0 +/- 8.0
Organs extraction fraction	%	19.0 +/- 5.0
Organs extravascular mean transit time	min	8.8 +/- 2.4
Gut mean transit time	sec	63.0 +/- 22.0
Gut dispersion	%	77.0 +/- 8.0
Liver extracellular volume fraction	%	20.0 +/- 2.0

Table 2.5: Constants for the drug rifampicin.

biomarker	units	value (95%CI)
Cardiac output	L/min	6.0 +/- 1.3
Heart-lung mean transit time	sec	17.0 +/- 2.5
Heart-lung dispersion	%	47.0 +/- 9.0
Organs blood mean transit time	sec	34.0 +/- 6.0
Organs extraction fraction	%	19.0 +/- 2.0
Organs extravascular mean transit time	min	8.4 +/- 2.6
Gut mean transit time	sec	38.0 +/- 5.4
Gut dispersion	%	83.0 +/- 9.0
Liver extracellular volume fraction	%	26.0 +/- 4.0

Table 2.6: Constants for the drug ciclosporin.

biomarker	units	value (95%CI)
Cardiac output	L/min	7.1 +/- 3.3
Heart-lung mean transit time	sec	20.0 +/- 1.9
Heart-lung dispersion	%	45.0 +/- 7.0
Organs blood mean transit time	sec	34.0 +/- 8.0
Organs extraction fraction	%	22.0 +/- 3.0
Organs extravascular mean transit time	min	9.1 +/- 1.5
Gut mean transit time	sec	34.0 +/- 10.3
Gut dispersion	%	83.0 +/- 7.0
Liver extracellular volume fraction	%	25.0 +/- 5.0

Table 2.7: Constants for the drug metformin.

biomarker	units	value (95%CI)
Cardiac output	L/min	6.8 +/- 0.2
Heart-lung mean transit time	sec	19.0 +/- 4.3
Heart-lung dispersion	%	41.0 +/- 10.0
Organs blood mean transit time	sec	29.0 +/- 3.0
Organs extraction fraction	%	22.0 +/- 2.0
Organs extravascular mean transit time	min	6.4 +/- 3.4
Gut mean transit time	sec	59.0 +/- 17.0
Gut dispersion	%	82.0 +/- 8.0
Liver extracellular volume fraction	%	24.0 +/- 7.0

Table 2.8: Constants for the drug rifampicin_clinical.

2.3. Effect plots

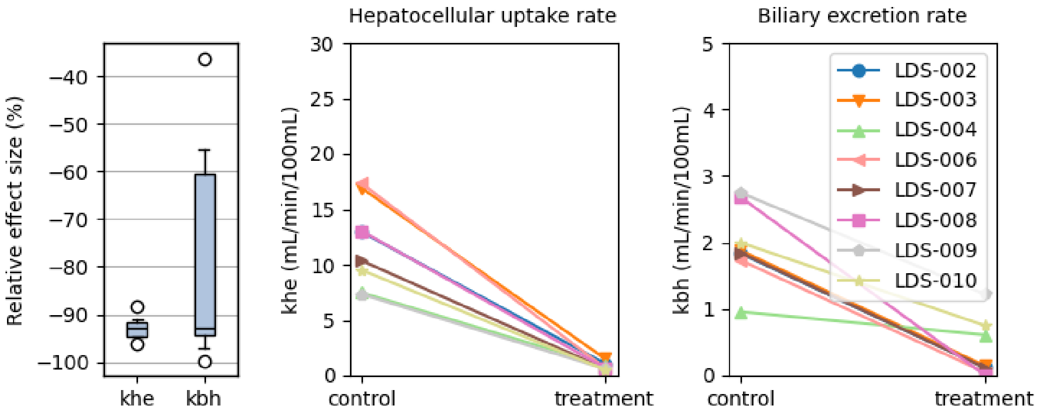

Figure 2.1: Effect of the drug rifampicin

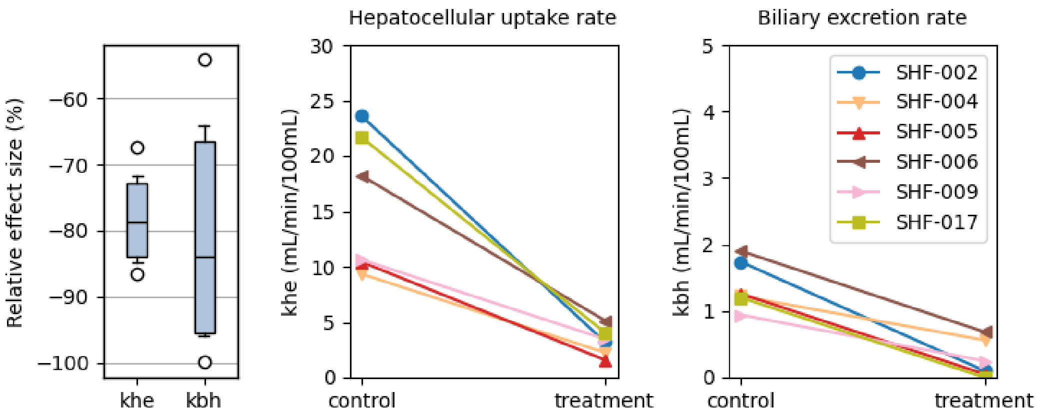

Figure 2.2: Effect of the drug ciclosporin

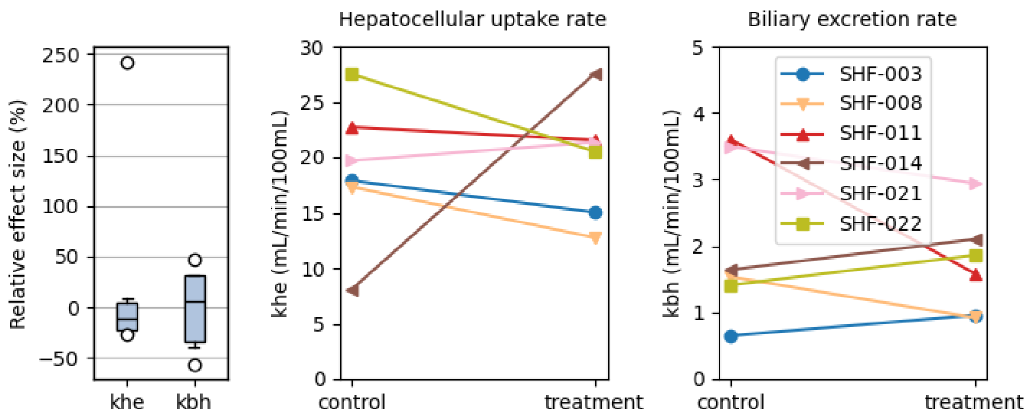

Figure 2.3: Effect of the drug metformin

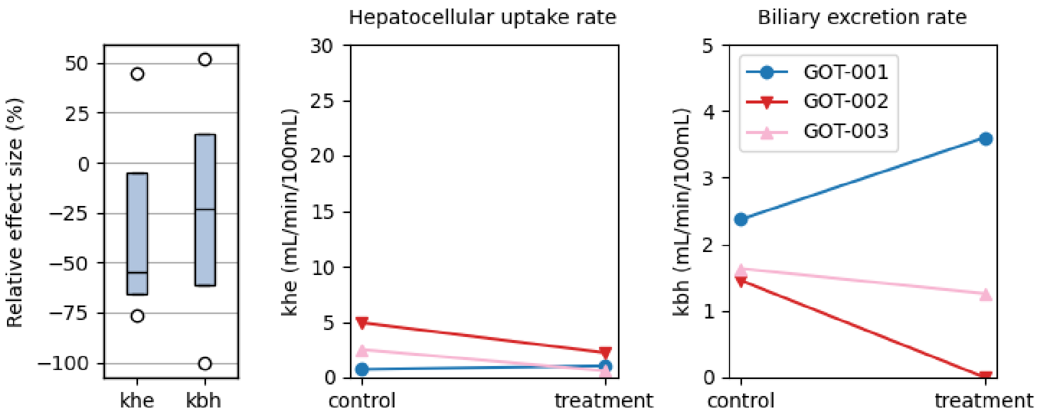

Figure 2.4: Effect of the drug rifampicin_clinical

2.4. Model fits

2.4.1. rifampicin

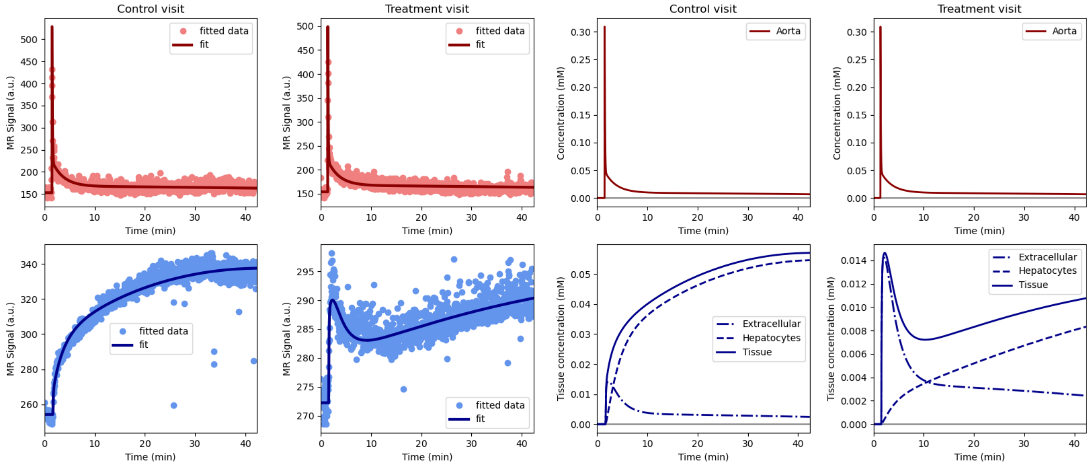

Figure 2.5: Volunteer LDS-002.png, drug rifampicin

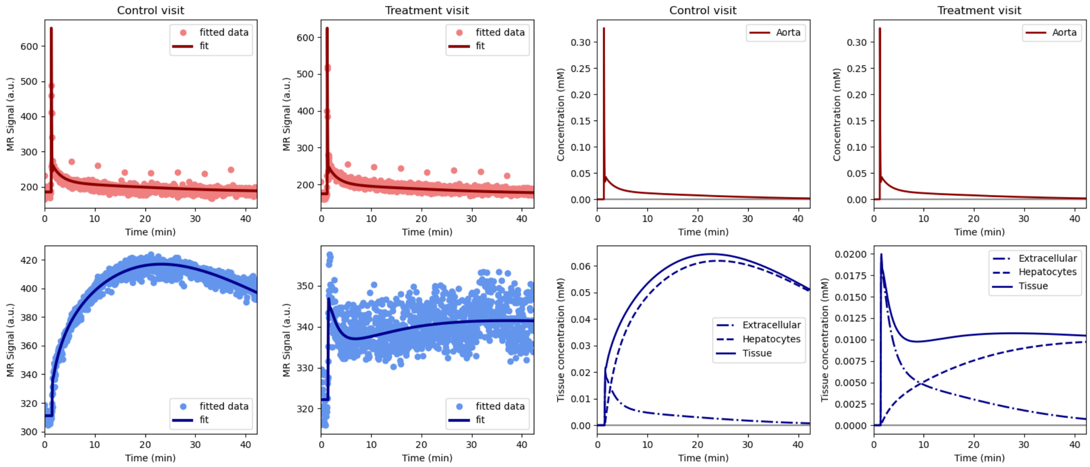

Figure 2.6: Volunteer LDS-003.png, drug rifampicin

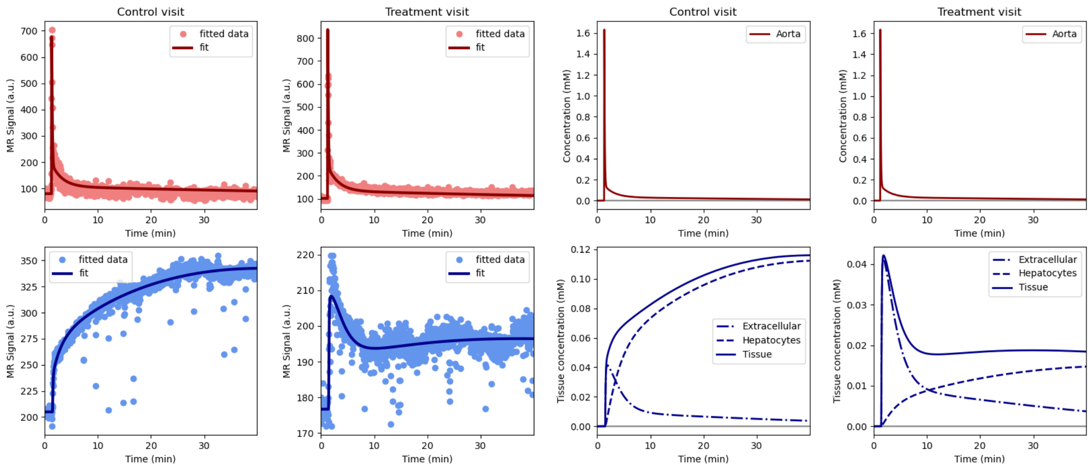

Figure 2.7: Volunteer LDS-004.png, drug rifampicin

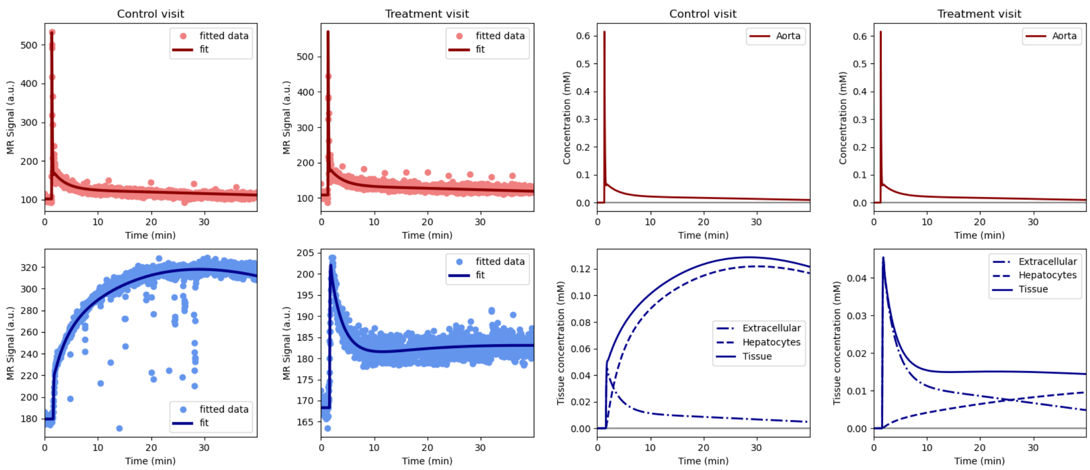

Figure 2.8: Volunteer LDS-006.png, drug rifampicin

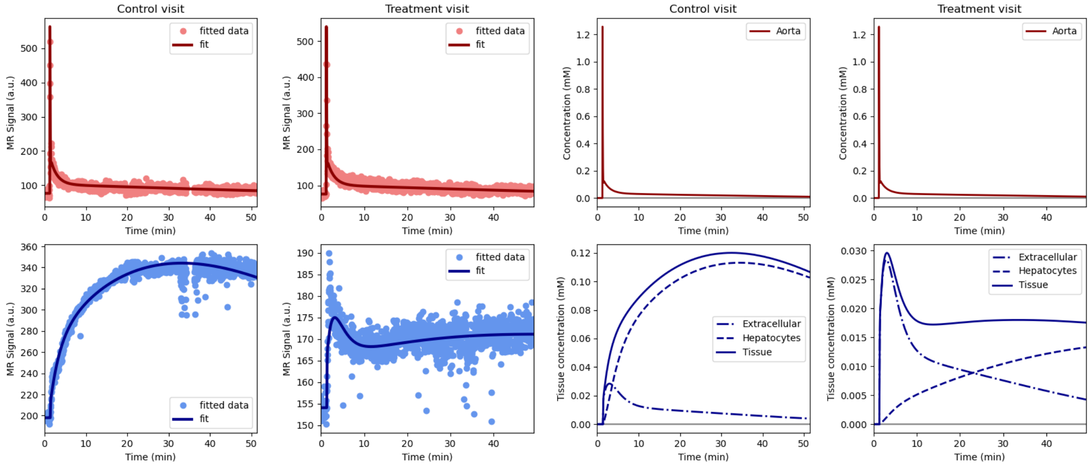

Figure 2.9: Volunteer LDS-007.png, drug rifampicin

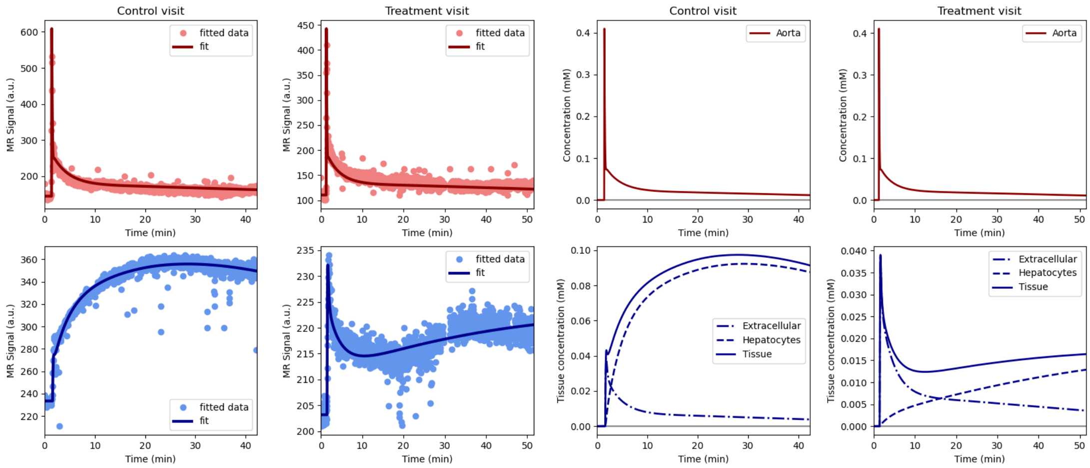

Figure 2.10: Volunteer LDS-008.png, drug rifampicin

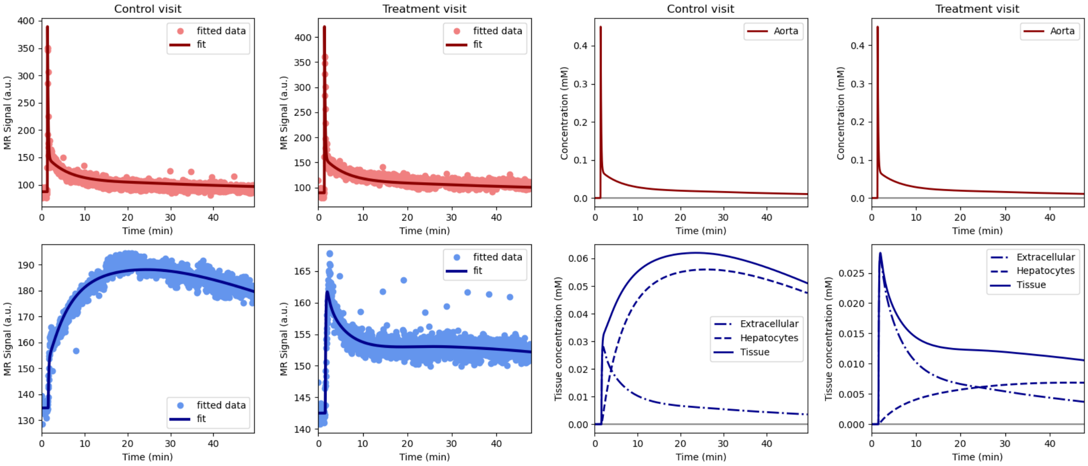

Figure 2.11: Volunteer LDS-009.png, drug rifampicin

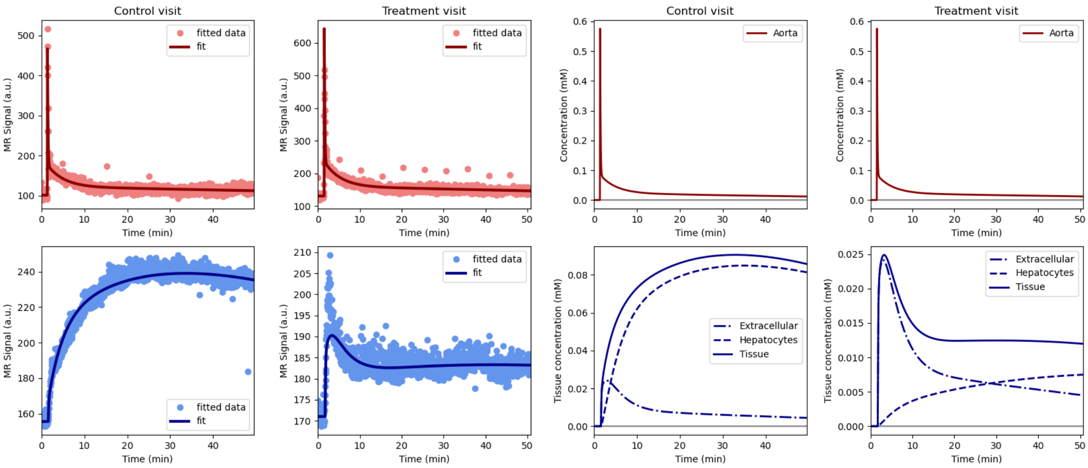

Figure 2.12: Volunteer LDS-010.png, drug rifampicin

2.4.2. ciclosporin

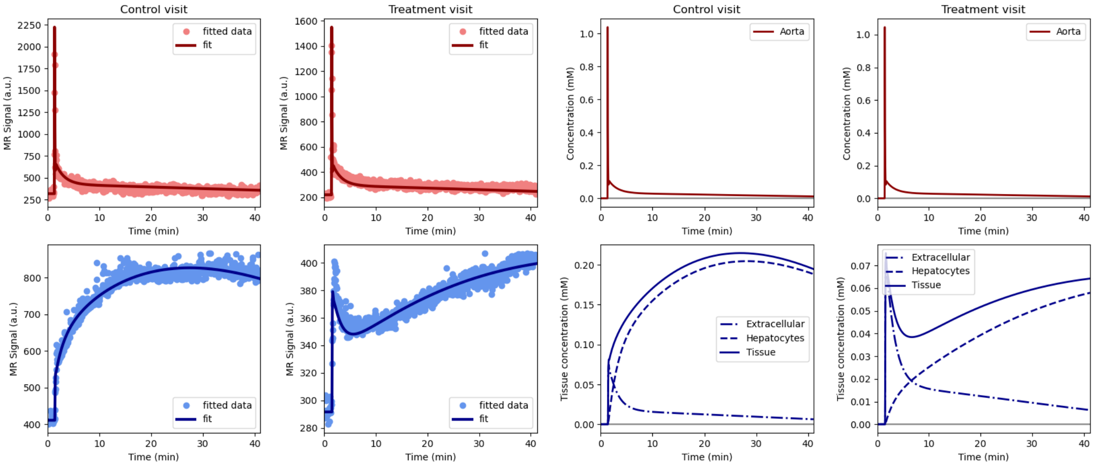

Figure 2.13: Volunteer SHF-002.png, drug ciclosporin

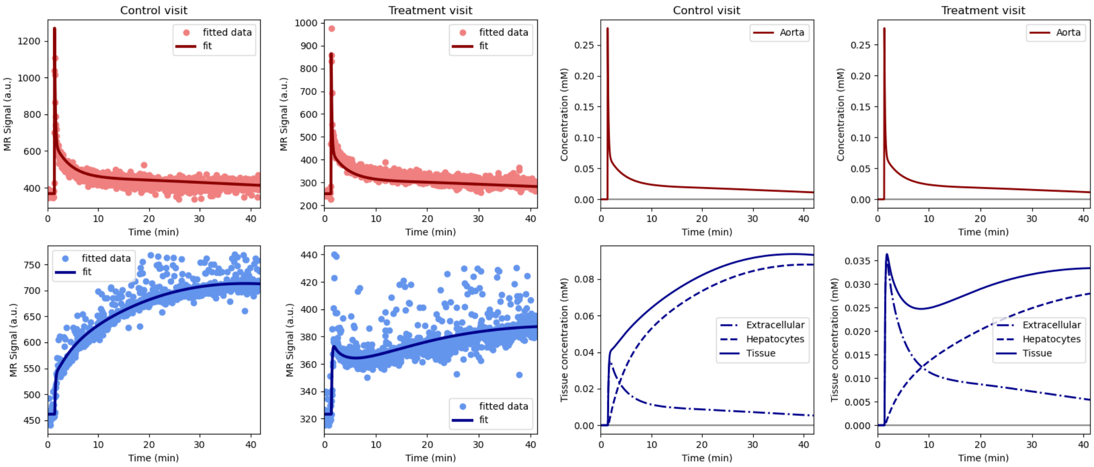

Figure 2.14: Volunteer SHF-004.png, drug ciclosporin

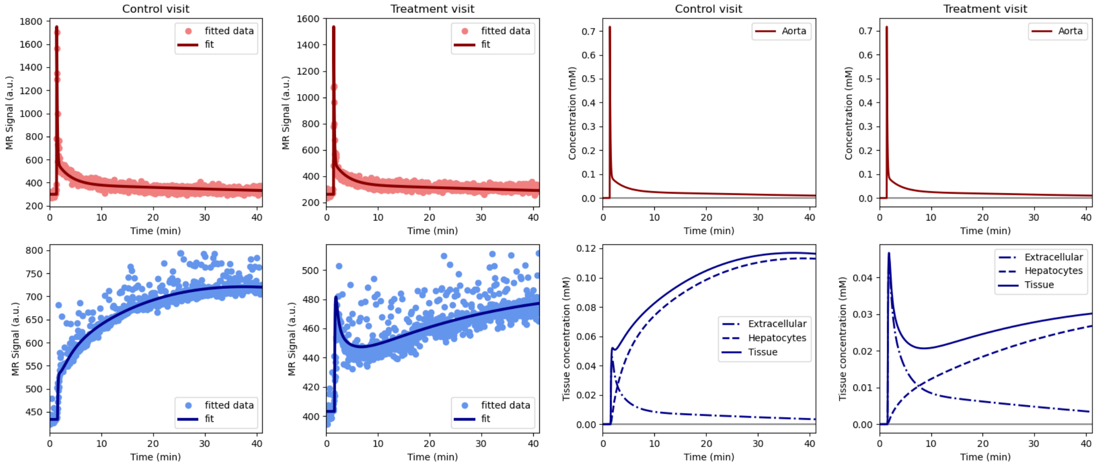

Figure 2.15: Volunteer SHF-005.png, drug ciclosporin

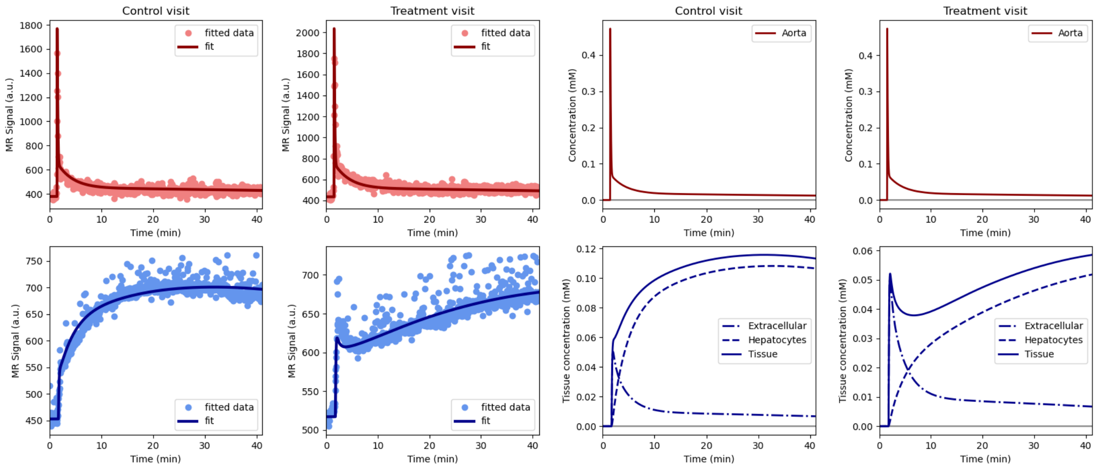

Figure 2.16: Volunteer SHF-006.png, drug ciclosporin

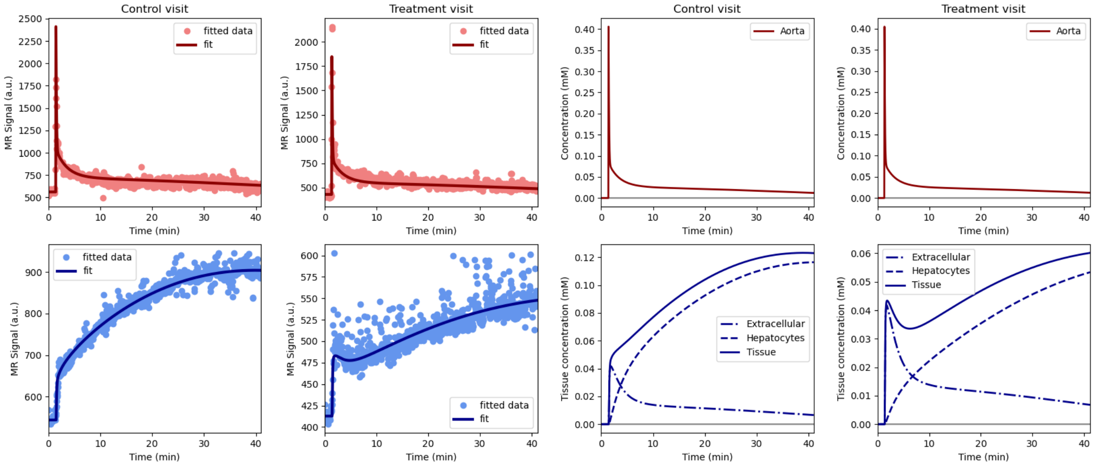

Figure 2.17: Volunteer SHF-009.png, drug ciclosporin

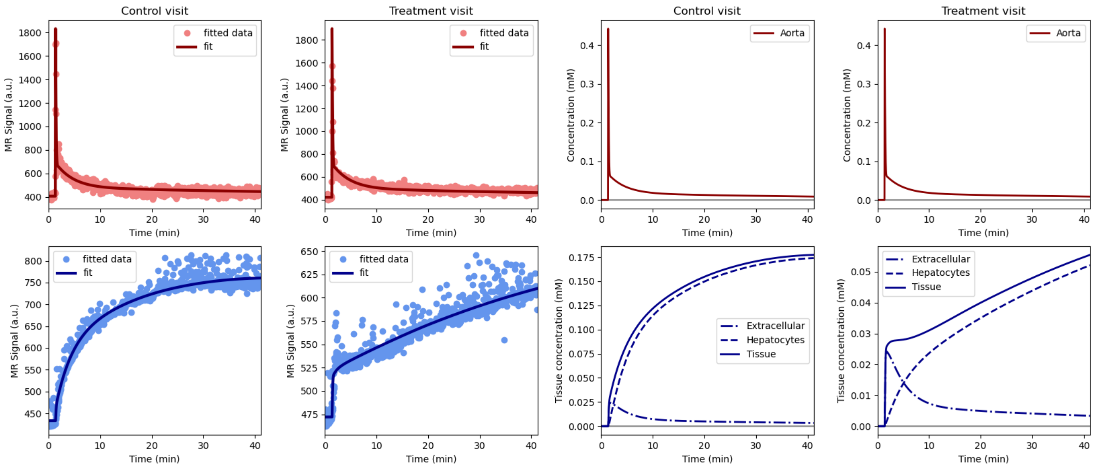

Figure 2.18: Volunteer SHF-017.png, drug ciclosporin

2.4.3. metformin

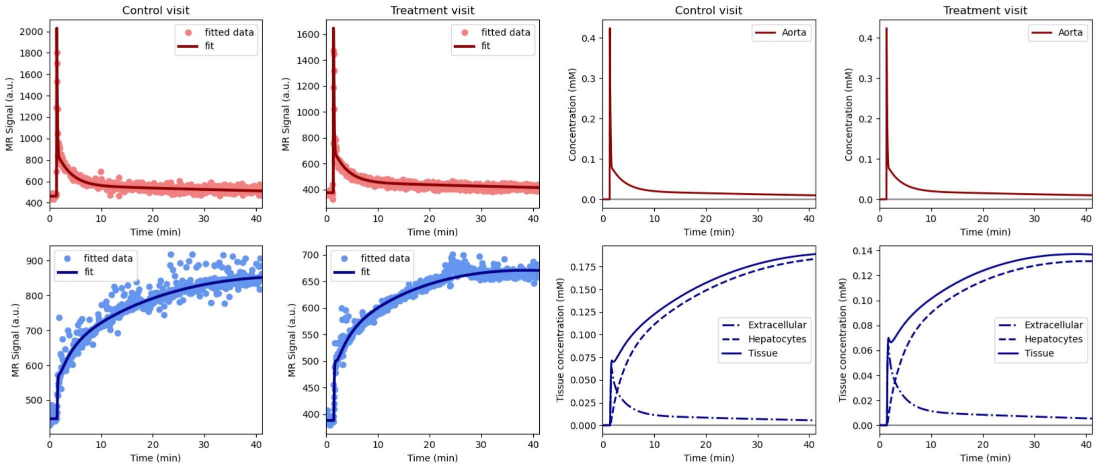

Figure 2.19: Volunteer SHF-003.png, drug metformin

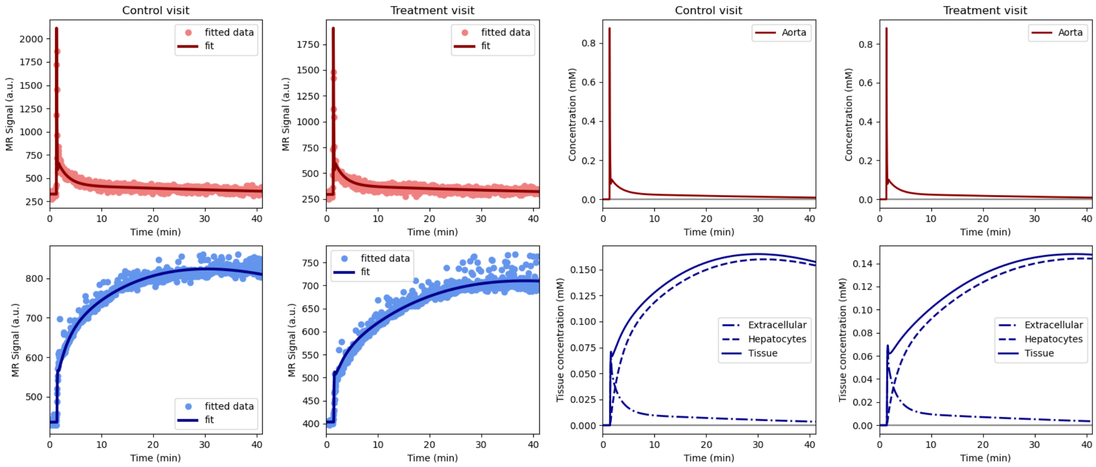

Figure 2.20: Volunteer SHF-008.png, drug metformin

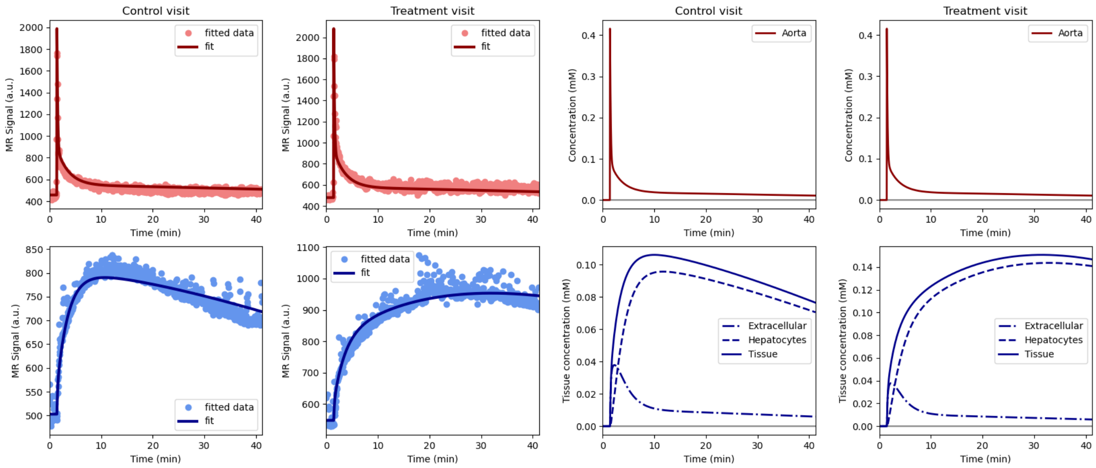

Figure 2.21: Volunteer SHF-011.png, drug metformin

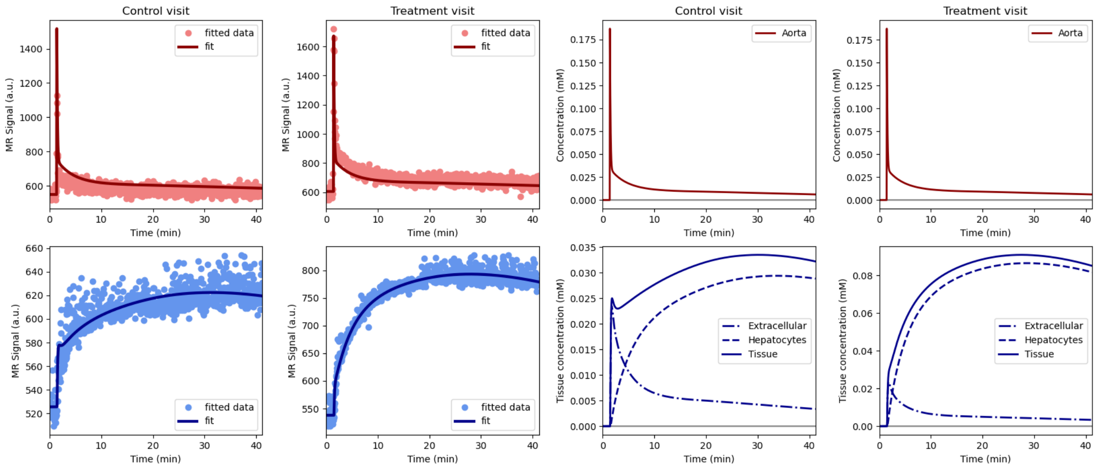

Figure 2.22: Volunteer SHF-014.png, drug metformin

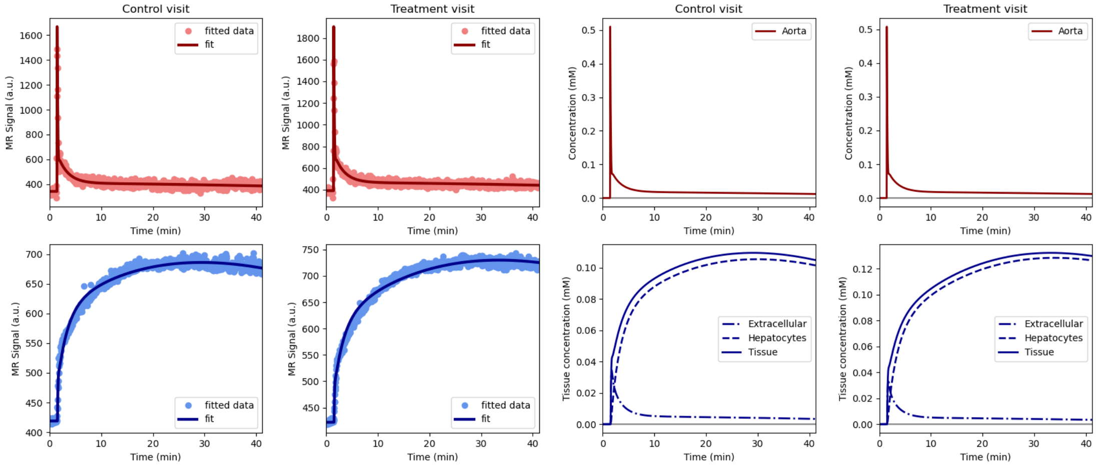

Figure 2.23: Volunteer SHF-021.png, drug metformin

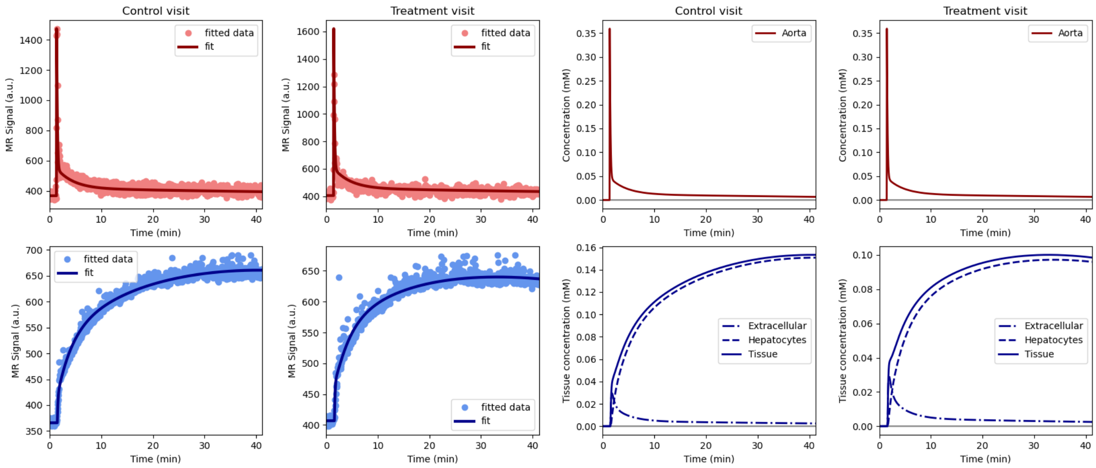

Figure 2.24: Volunteer SHF-022.png, drug metformin

2.4.4. rifampicin_clinical

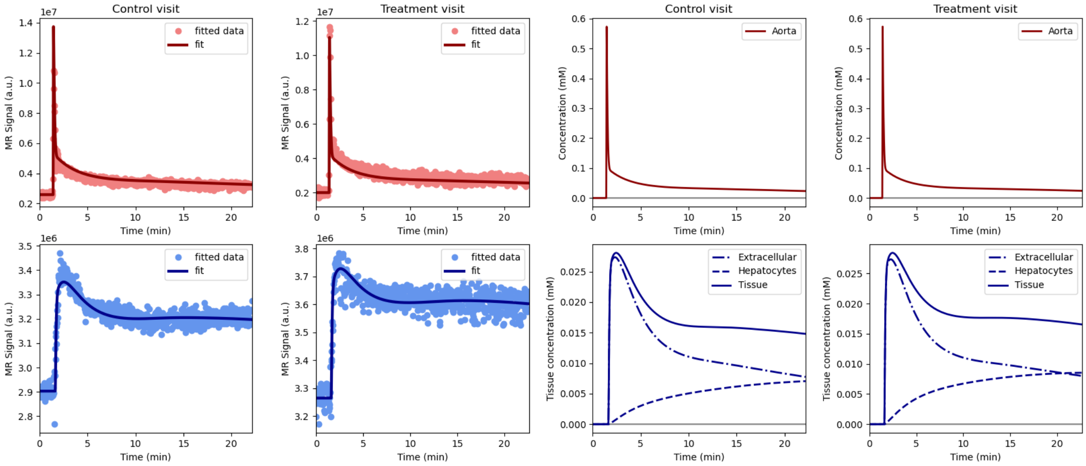

Figure 2.25: Volunteer GOT-001.png, drug rifampicin_clinical

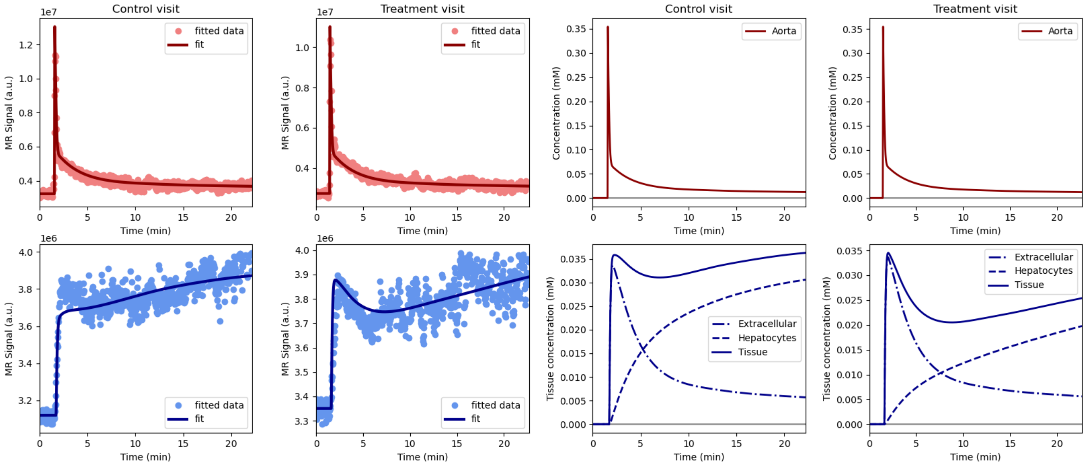

Figure 2.26: Volunteer GOT-002.png, drug rifampicin_clinical

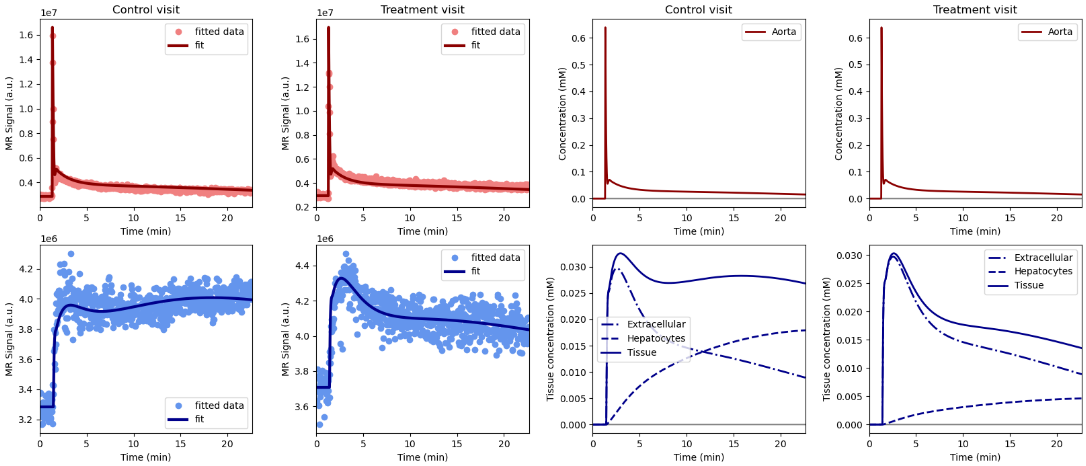

Figure 2.27: Volunteer GOT-003.png, drug rifampicin_clinical